

Unit I

Introduction to Networks



TOPICS

- Introduction & History
- Network Types – LAN, MAN, WAN, PAN
- Network Topology – BUS, STAR, RING, MESH & HYBRID
- Switching – Circuit & Packet switching
- OSI Layered Architecture
- TCP/IP Model
- Physical Layer Overview – Latency, B/W etc.
- Guided Media – TP, Coaxial & Fiber Optic
- Unguided Media – Radio waves, Microwave & Infrared

Resources:

Behrouz A. Forouzan, “Data Communication & Networking”, 5th edition, 2010.
Bhusan Trivedi, Data Communication & Networks”, 2016.

Introduction & Basic Concepts

Definition:

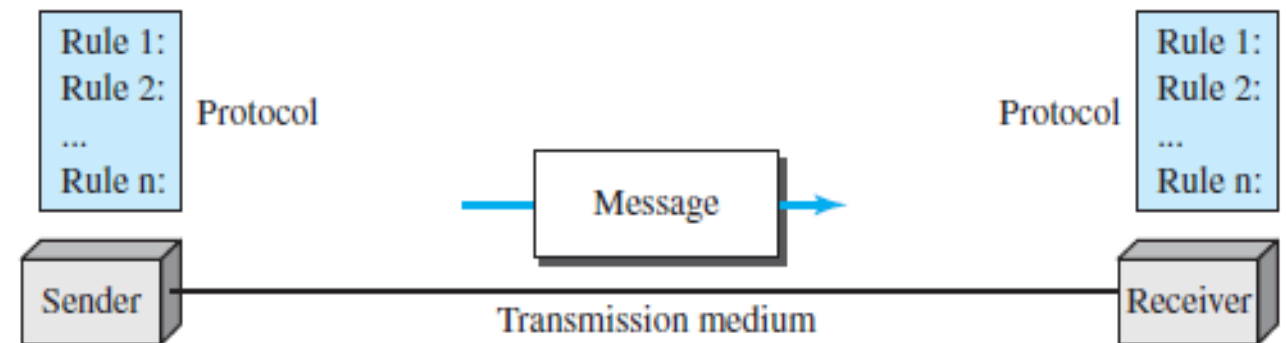
- A computer network consists of two or more computing devices connected to each other to share resources and information.
- The network becomes a powerful tool when computers communicate and share resources with other computers on the same network or entirely distinct networks.
- Computers on a network can act as a client or a server.
- A client is a computer that requests for resources.
- A server is a computer that controls and provides access to resources.
- The word ***data*** refers to information presented in whatever form is agreed upon by the parties creating and using the data.
- **Data communications** are the exchange of data between two devices via some form of transmission medium such as a wire cable.

Introduction to Networks (Contd.)

- For data communications to occur, the communicating devices must be part of a communication system made up of a combination of hardware (physical equipment) and software (programs).
- The computing concept 'hierarchy of data' is used when planning a network.
- It is essential to maintain a hierarchy of data to manage and control resources among computers.
- Network access to data must be evaluated carefully to avoid security issues

1. Protocol
2. Sender
3. Message
4. Receiver
5. Tx Medium (Wired/Wireless)

Five components of data communication



Need for Networks

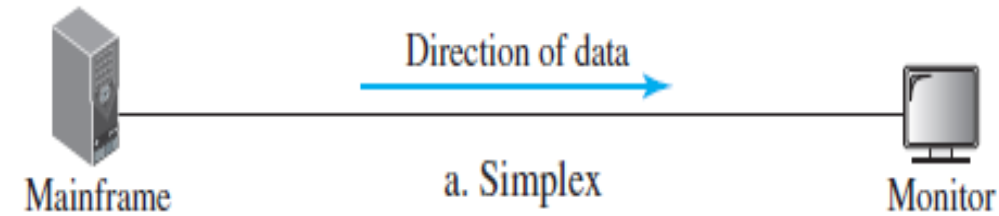
- **Enhance communication**
 - By using networks, information can be sent to a larger audience in an extremely fast and efficient manner.
- **Share resources.**
 - A copy of data or application stored at a single central location is shared over a network. Hardware resource can also be shared.
- **Facilitate centralized management.**
 - Networks are used to assist in management tasks associated with their own operation and maintenance.
 - Using networks results in increased efficiency and a resultant reduction in maintenance costs.

Data Flow

- Communication between two devices can be simplex, half-duplex, or full-duplex as shown in Figure.

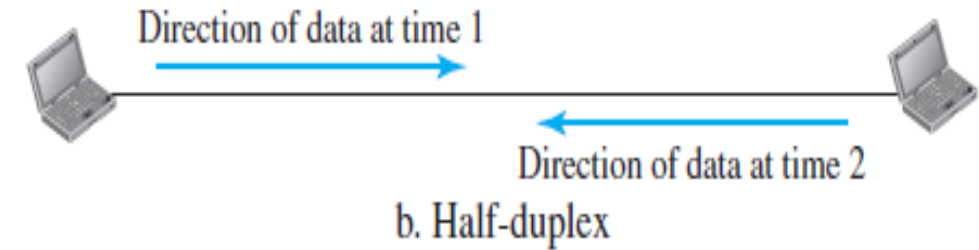
a) Simplex

- In **simplex mode**, the communication is unidirectional, as on a one-way street. Only one of the two devices on a link can transmit; the other can only receive. Keyboards and traditional monitors are examples of simplex devices. The simplex mode can use the entire capacity of the channel to send data in one direction.



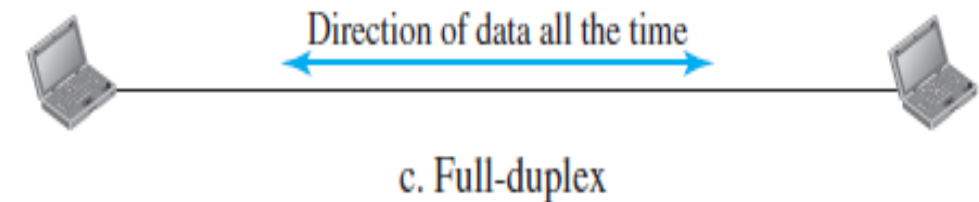
b) Half-Duplex

- In **half-duplex mode**, each station can both transmit and receive, but not at the same time. When one device is sending, the other can only receive, and vice versa. Walkie-talkies and CB (citizens band) radios are both half-duplex systems.



c) Full-Duplex

- In **full-duplex mode** (also called *duplex*), both stations can transmit and receive simultaneously. The full-duplex mode is like a two-way street with traffic flowing in both directions at the same time. One common example of full-duplex communication is the telephone network.



Network & It's Criteria

- A **network** is the interconnection of a set of devices capable of communication. In this definition, a device can be a **host** (or an *end system* as it is sometimes called) such as a large computer, desktop, laptop, workstation, cellular phone, or security system. A device in this definition can also be a **connecting device** such as a router, which connects the network to other networks, a switch, which connects devices together, a modem (modulator-demodulator), which changes the form of data, and so on. These devices in a network are connected using wired or wireless transmission media such as cable or air. When we connect two computers at home using a plug-and-play router, we have created a network, although very small.
- A network must be able to meet a certain number of criteria. The most important of these are **performance**, **reliability**, and **security**.

Classification of Networks

- Classification by Network Geography (**LAN, MAN, WAN, PAN**).
- Classification by Physical Topology (**Star, Bus, Ring, Mesh**).

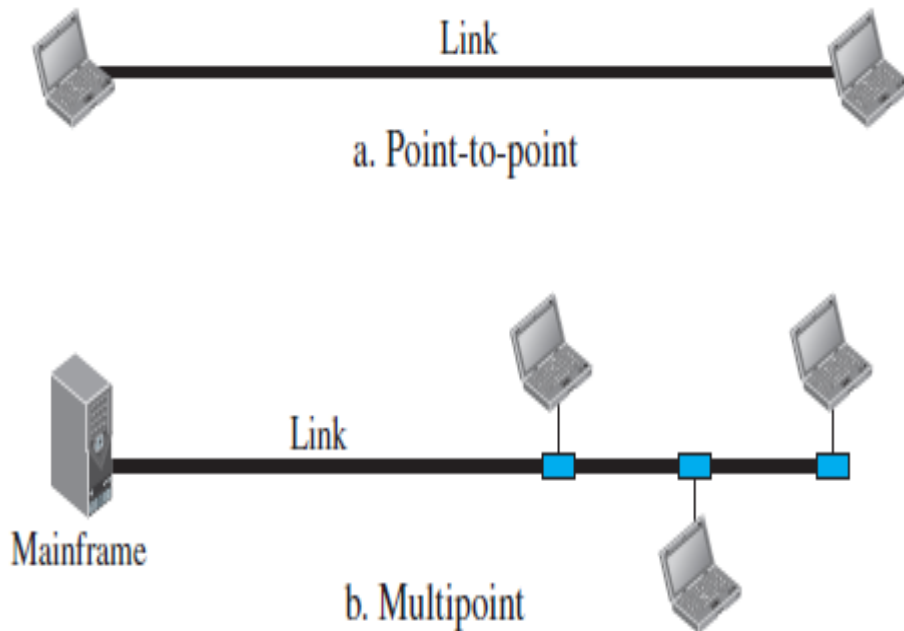
Before discussing networks, we need to define some network attributes.

Point-to-Point

A **point-to-point connection** provides a dedicated link between two devices. The entire capacity of the link is reserved for transmission between those two devices. Most point-to-point connections use an actual length of wire or cable to connect the two ends, but other options, such as microwave or satellite links, are also possible. When we change television channels by infrared remote control, we are establishing a point-to-point connection between the remote control and the television's control system.

Multipoint

A **multipoint** (also called **multidrop**) **connection** is one in which more than two specific devices share a single link. In a multipoint environment, the capacity of the channel is shared, either spatially or temporally. If several devices can use the link simultaneously, it is a *spatially shared* connection. If users must take turns, it is a *timeshared* connection.



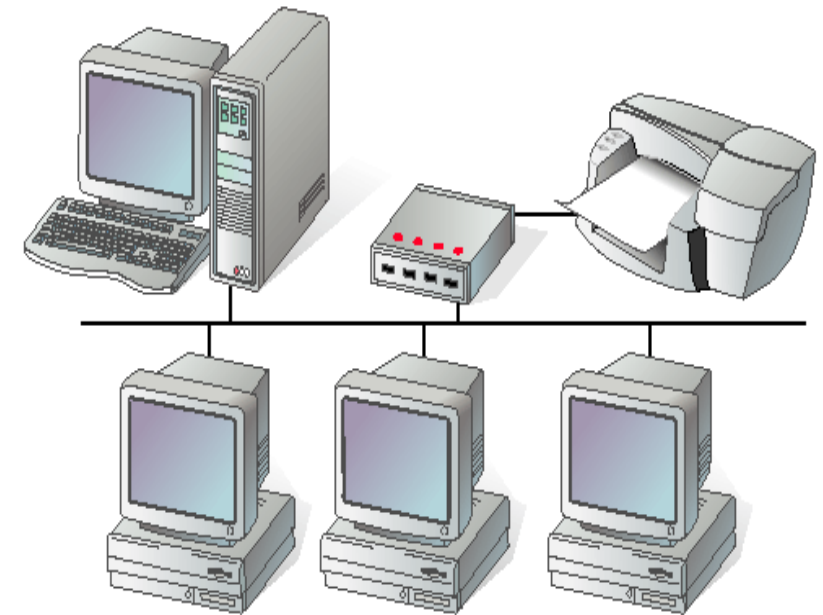
Classification by Network Geography

- Networks are frequently classified according to the geographical boundaries spanned by the network itself.
- LAN, WAN, PAN and MAN are the basic types of classification, of which **LAN** and **WAN** are frequently used.

1. LAN

A **local area network (LAN)** is usually privately owned and connects some hosts in a single office, building, or campus. Depending on the needs of an organization, a LAN can be as simple as two PCs and a printer in someone's home office, or it can extend throughout a company and include audio and video devices.

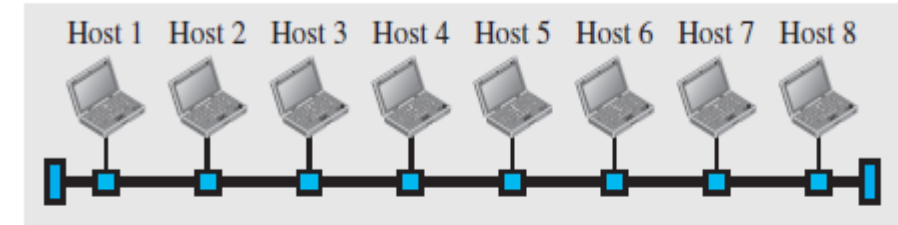
LANs are inexpensive to install and also provide higher speeds
LAN also provides higher security.



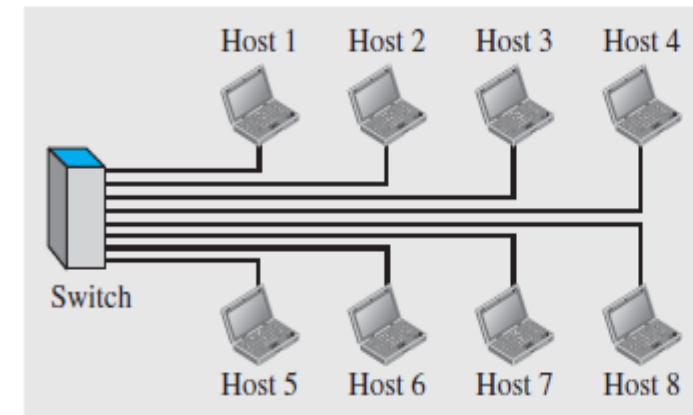
LAN

In the past, all hosts in a network were connected through a common cable, which meant that a packet sent from one host to another was received by all hosts. The intended recipient kept the packet; the others dropped the packet.

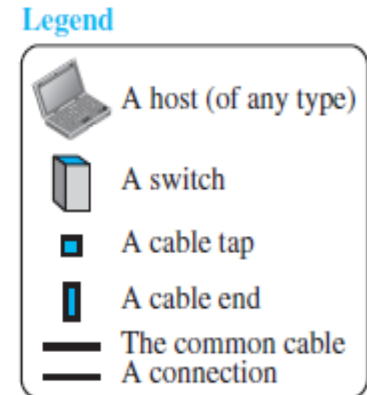
Today, most LANs use a smart connecting switch, which is able to recognize the destination address of the packet and guide the packet to its destination without sending it to all other hosts. The switch alleviates the traffic in the LAN and allows more than one pair to communicate with each other at the same time if there is no common source and destination among them.



a. LAN with a common cable (past)



b. LAN with a switch (today)

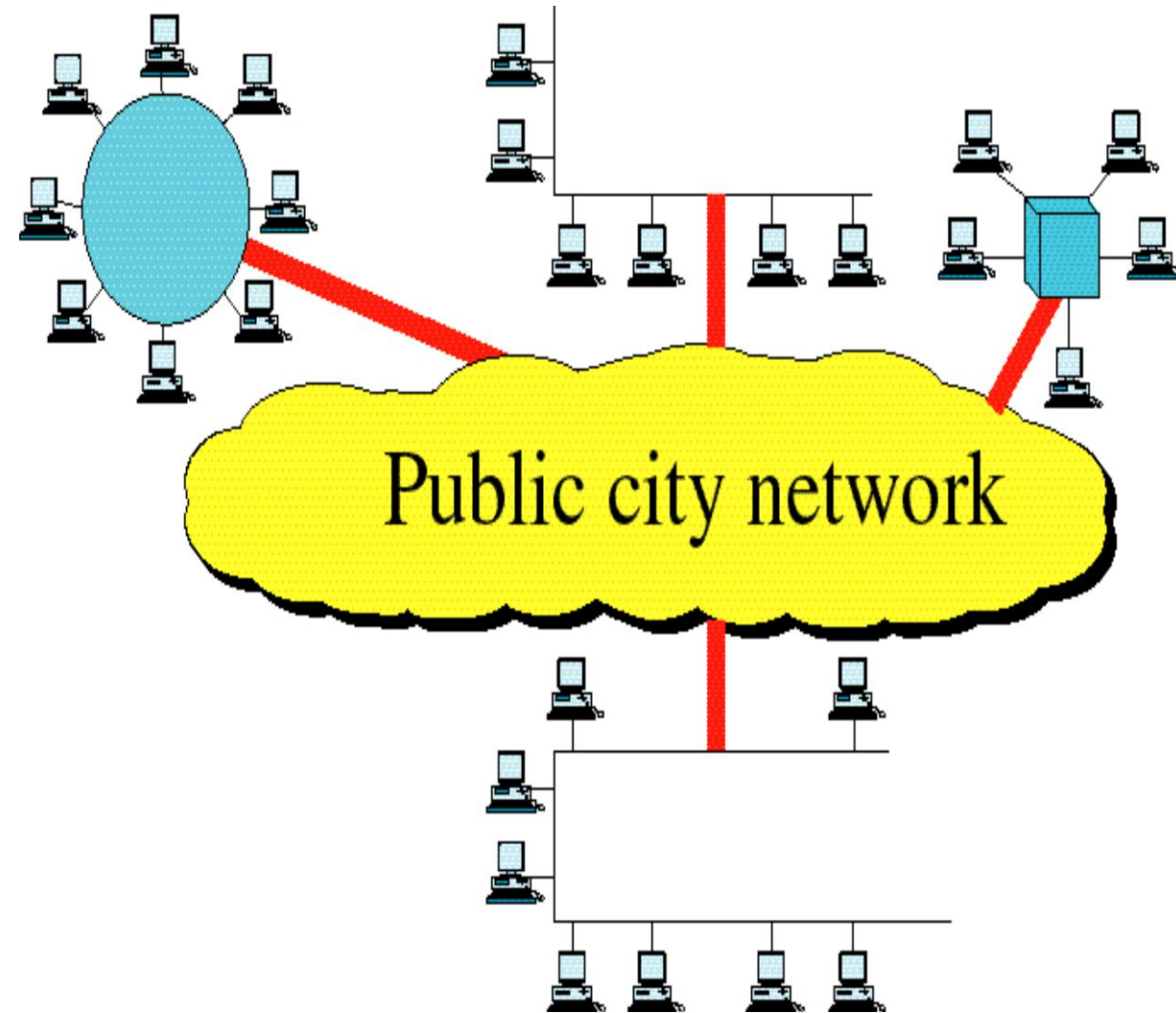


Characteristics of LAN:

- LAN's are private networks, not subject to tariffs or other regulatory controls.
- LAN's operate at relatively high speed when compared to the typical WAN.
- There are different types of Media Access Control methods in a LAN, the prominent ones are Ethernet, Token ring.
- It connects computers in a single building, block or campus, i.e. they work in a restricted geographical area.

Metropolitan Area Network

- A metropolitan area network is a network that covers a larger geographic area by interconnecting a different LAN to form a larger network.
- Government agencies use MAN to connect to the citizens and private industries.
- In MAN, various LANs are connected to each other through a telephone exchange line.
- The most widely used protocols in MAN are RS-232, Frame Relay, ATM, ISDN, OC-3, ADSL, etc.
- It has a higher range than Local Area Network(LAN).



MAN (Contd.)

Characteristics:

- It generally covers towns and cities (50 km)
- Communication medium used for MAN are optical fibers, cables etc.
- Data rates adequate for distributed computing applications.

Applications:

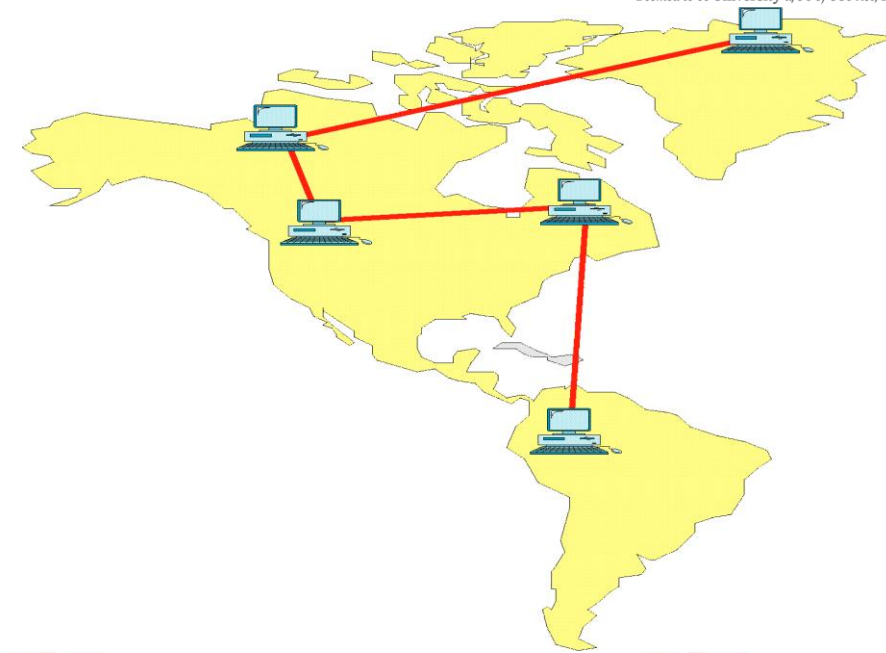
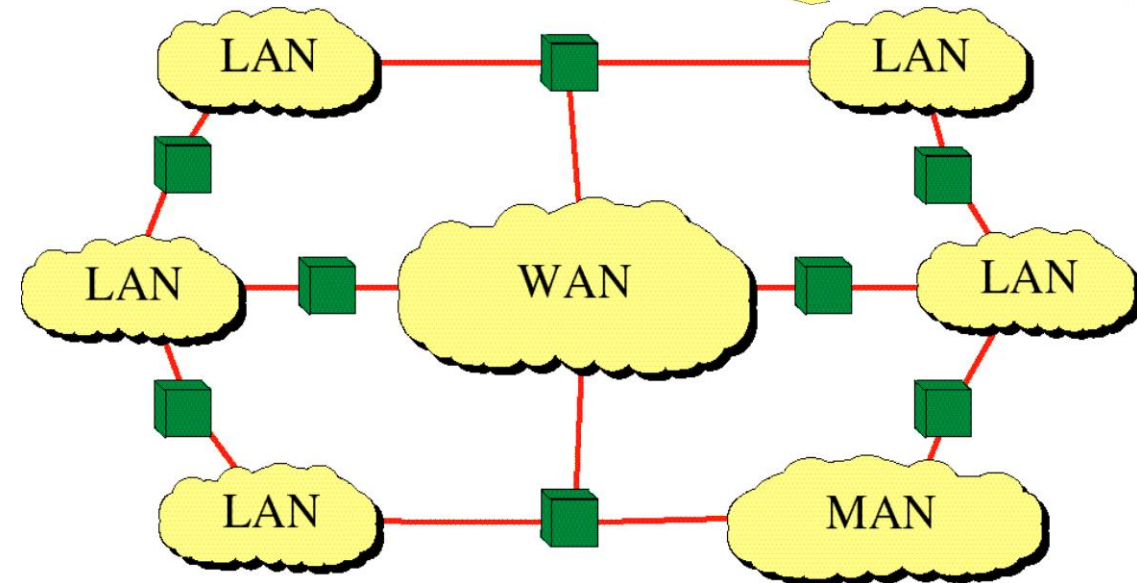
- MAN is used in communication between the banks in a city.
- It can be used in an Airline Reservation.
- It can be used in a college campuses within a city.

Wide Area Network (WAN)

- WAN can be private, or it can be public leased network.
- It is used for the network that covers large distance such as cover states of a country.
- It is not easy to design and maintain.
- Communication medium used by WAN are PSTN or Satellite links.
- WAN operates on low data rates.

Example:

- **Mobile Broadband:** A 4G network is widely used across a region or country.
- **Private network:** A bank provides a private network that connects the 44 offices. This network is made by using the telephone leased line provided by the telecom company.



WAN (Contd.)

Characteristics

- It generally covers large distances(states, countries, continents).
- Communication medium used are satellite, public telephone networks which are connected by routers.

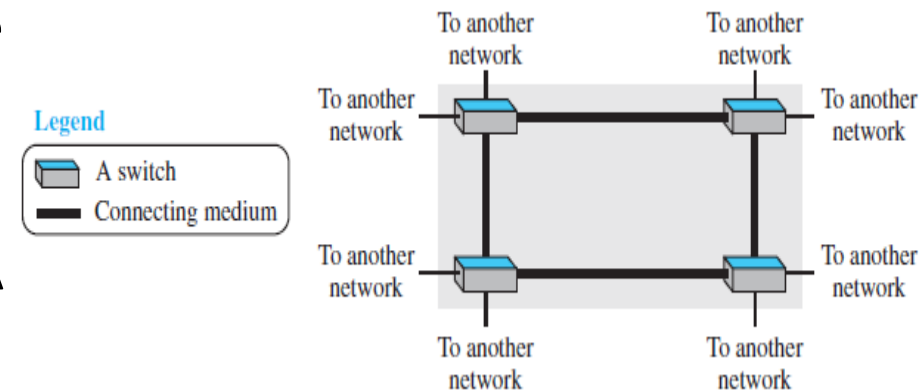
Point-to-Point WAN

- A point-to-point WAN is a network that connects two communicating devices through a transmission media (cable or air).



Switched WAN

- A switched WAN is a network with more than two ends. A switched WAN is used in the backbone of global communication today. We can say that a switched WAN is a combination of several point-to-point WANs that are connected by switches.



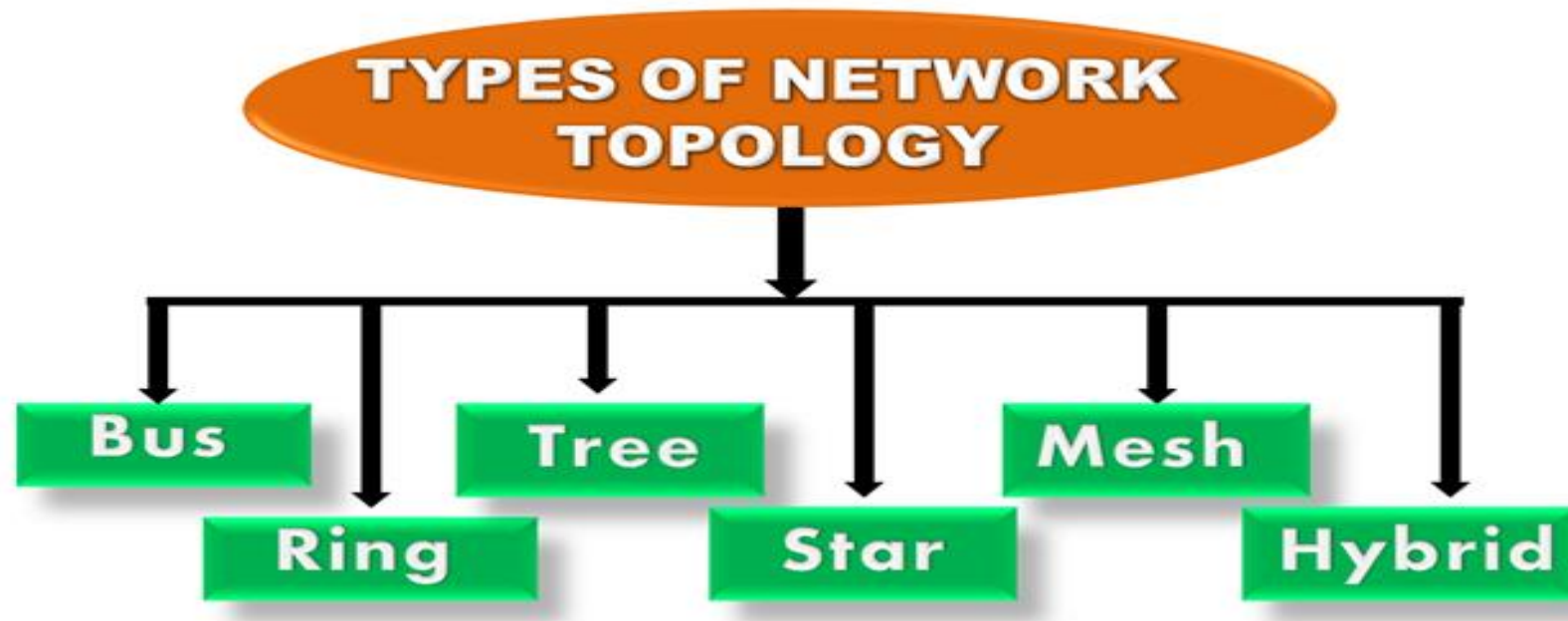
Personal Area Network (PAN)

- **Thomas Zimmerman** was the first research scientist to bring the idea of the Personal Area Network.
- Personal Area Network is a network arranged within an individual person, typically within a range of **10 meters (30 feet)**.
- Personal Area Network is used for connecting the computer devices of personal use is known as Personal Area Network.
- Personal computer devices that are used to develop the personal area network are the laptop, mobile phones, media player and play stations.

Example: Body Area Network: Body Area Network is a network that moves with a person. **For example**, a mobile network moves with a person.



Classification by Physical Topology



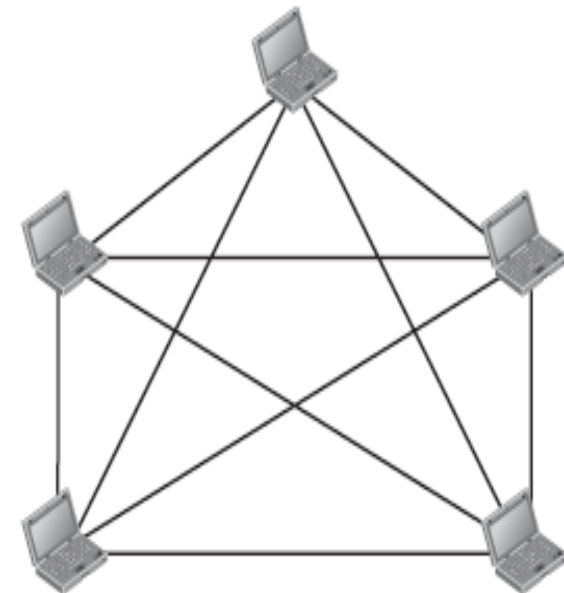
Mesh Topology

- In a **mesh topology**, every device has a dedicated point-to-point link to every other device. The term *dedicated* means that the link carries traffic only between the two devices it connects. To find the number of physical links in a fully connected mesh network with n nodes, we first consider that each node must be connected to every other node.
- In a mesh topology, we need $n(n - 1) / 2$ duplex-mode links.

Advantages:

1. First, the use of dedicated links guarantees that each connection can carry its own data load, thus eliminating the traffic problems that can occur when links must be shared by multiple devices.
2. Second, a mesh topology is robust. If one link becomes unusable, it does not incapacitate the entire system.
3. Third, there is the advantage of privacy or security. When every message travels along a dedicated line, only the intended recipient sees it. Physical boundaries prevent other users from gaining access to messages.
4. Finally, point-to-point links make fault identification and fault isolation easy.

$n = 5$
10 links.



Mesh Topology (Contd.)

Disadvantages:

The main disadvantages of a mesh are related to the amount of cabling and the number of I/O ports required.

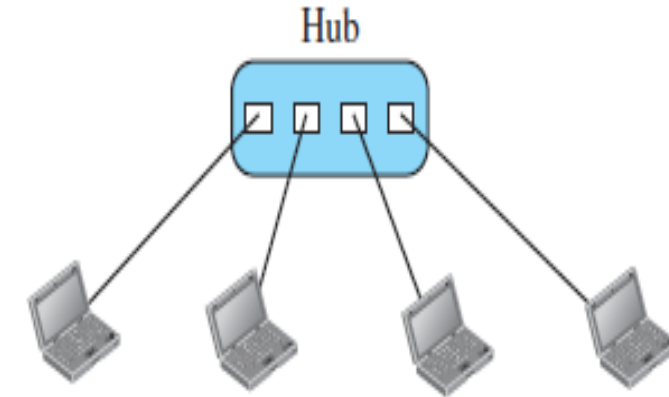
1. First, because every device must be connected to every other device, installation and reconnection are difficult.
2. Second, the sheer bulk of the wiring can be greater than the available space (in walls, ceilings, or floors) can accommodate.
3. Finally, the hardware required to connect each link (I/O ports and cable) can be prohibitively expensive.

Example:

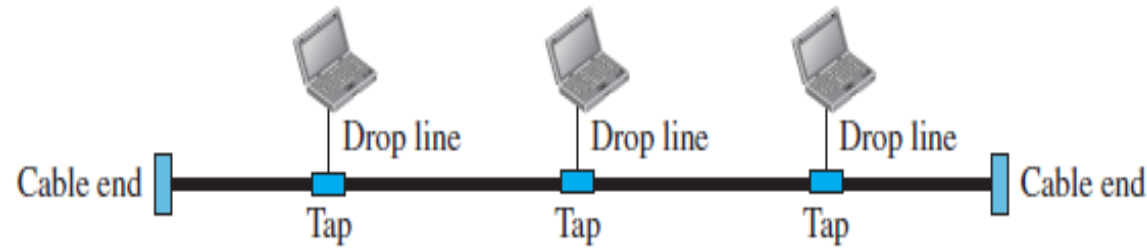
- One practical example of a mesh topology is the connection of telephone regional offices in which each regional office needs to be connected to every other regional office.

Star Topology

- In a **star topology**, each device has a dedicated point-to-point link only to a central controller, usually called a **hub**. The devices are not directly linked to one another.
- Unlike a mesh topology, a star topology does not allow direct traffic between devices. The controller acts as an exchange: If one device wants to send data to another, it sends the data to the controller, which then relays the data to the other connected device.
- A star topology is less expensive than a mesh topology. In a star, each device needs only one link and one I/O port to connect it to any number of others. This factor also makes it easy to install and reconfigure.
- Other advantages include robustness. If one link fails, only that link is affected.
- One big disadvantage of a star topology is the dependency of the whole topology on one single point, the hub. If the hub goes down, the whole system is dead.
- The star topology is used in local-area networks (LANs)



Bus Topology



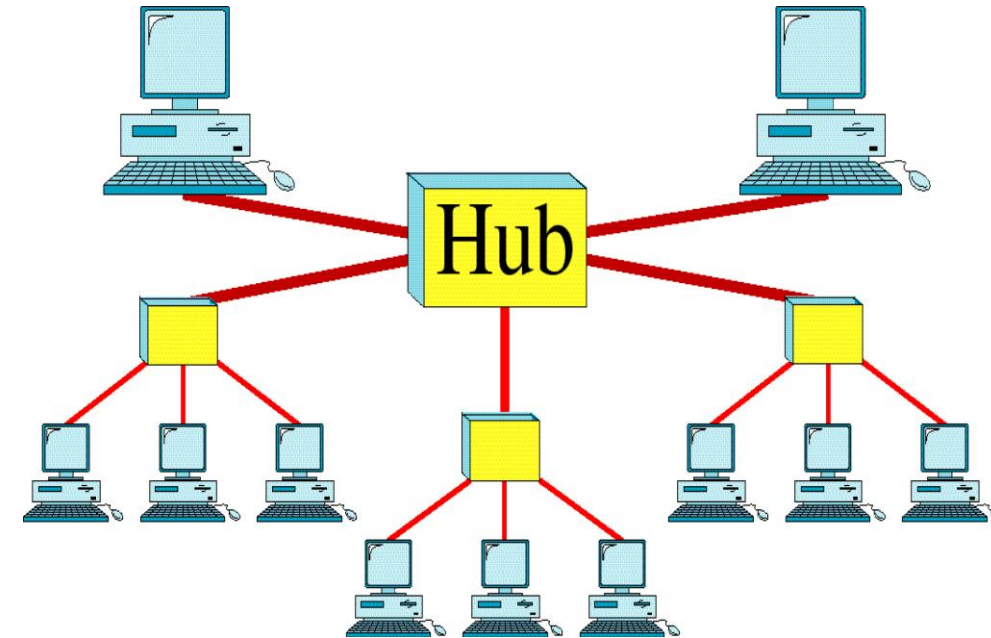
- A **bus topology** is based on multipoint connection. One long cable acts as a **backbone** to link all the devices in a network.
- Nodes are connected to the bus cable by drop lines and taps. A drop line is a connection running between the device and the main cable. A tap is a connector that either splices into the main cable or punctures the sheathing of a cable to create a contact with the metallic core.
- Advantages of a bus topology include ease of installation. Backbone cable can be laid along the most efficient path, then connected to the nodes by drop lines of various lengths. In this way, a bus uses less cabling than mesh or star topologies.
- Disadvantages include difficult reconnection and fault isolation. A bus is usually designed to be optimally efficient at installation. It can therefore be difficult to add new devices. Signal reflection at the taps can cause degradation in quality. In addition, a fault or break in the bus cable stops all transmission, even between devices on the same side of the problem.

Example:

- Traditional Ethernet LANs can use a bus topology, but they are less popular now.

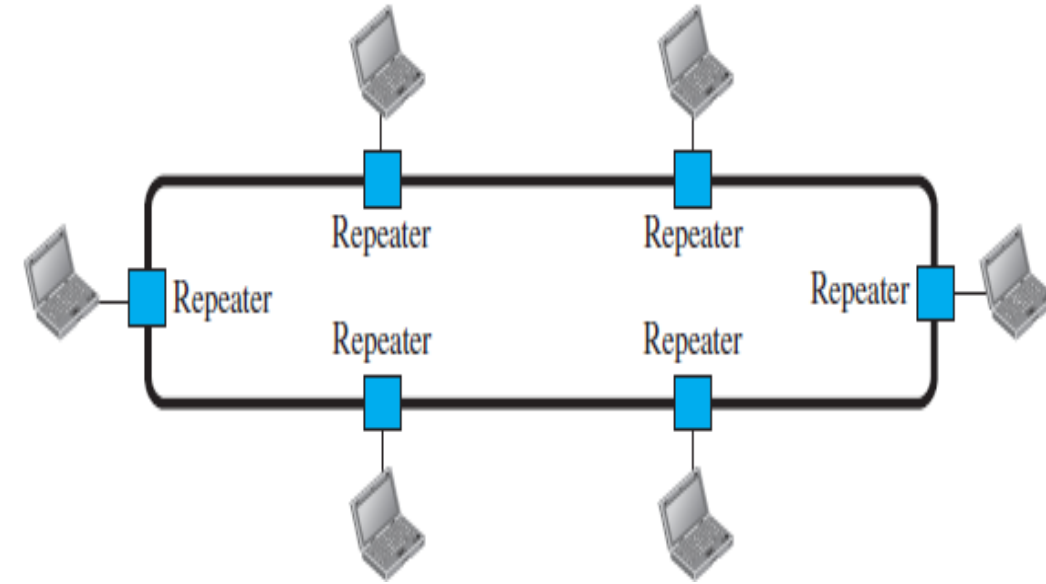
Tree Topology

- It's is a type of network topology that resembles a tree. In a tree topology, there is one central node (the “trunk”), and each node is connected to the central node through a single path.
- Sometimes it is also called **hierarchical topology** as in this topology, all elements are arranged like the branches of a tree.
- Tree topologies are commonly used to **arrange data** in databases and **workstations** in corporate networks. In a tree topology, any two linked nodes can only have one mutual connection, hence there can only be one link between them.
- The disadvantage of a tree topology is that if the primary node is damaged, the entire system can be crippled, because all other nodes are connected to the primary node.



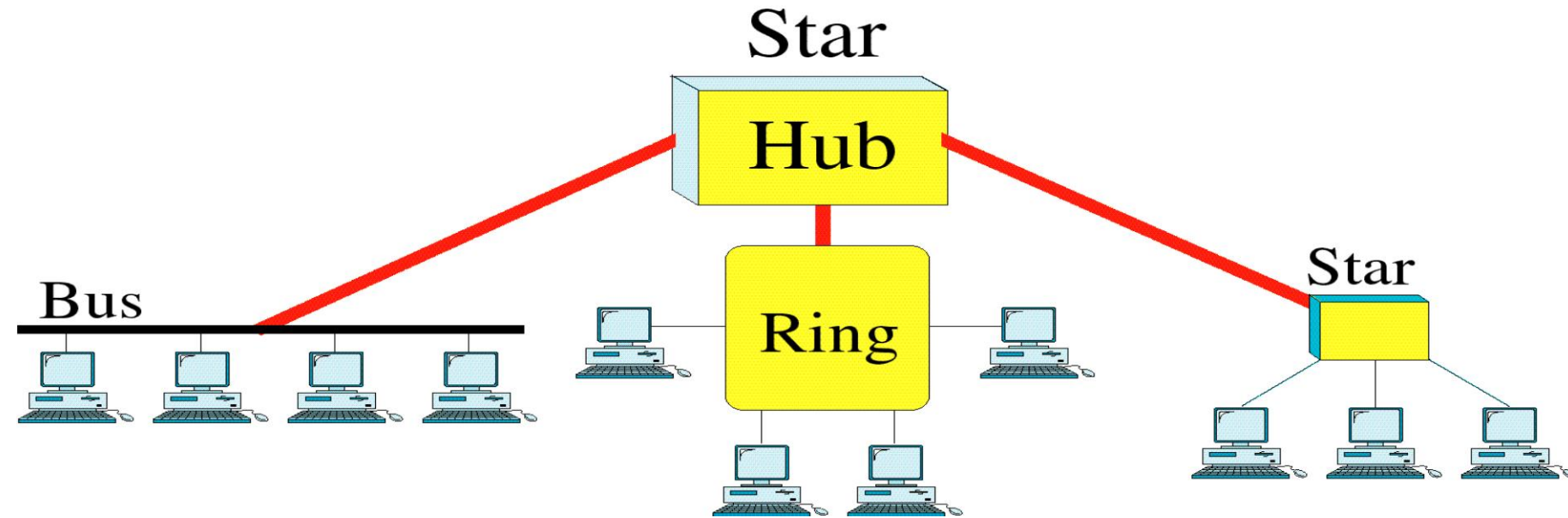
Ring Topology

- In a **ring topology**, each device has a dedicated point-to-point connection with only the two devices on either side of it.
- A signal is passed along the ring in one direction, from device to device, until it reaches its destination. Each device in the ring incorporates a repeater.
- A ring is relatively easy to install and reconfigure. Each device is linked to only its immediate neighbors (either physically or logically).



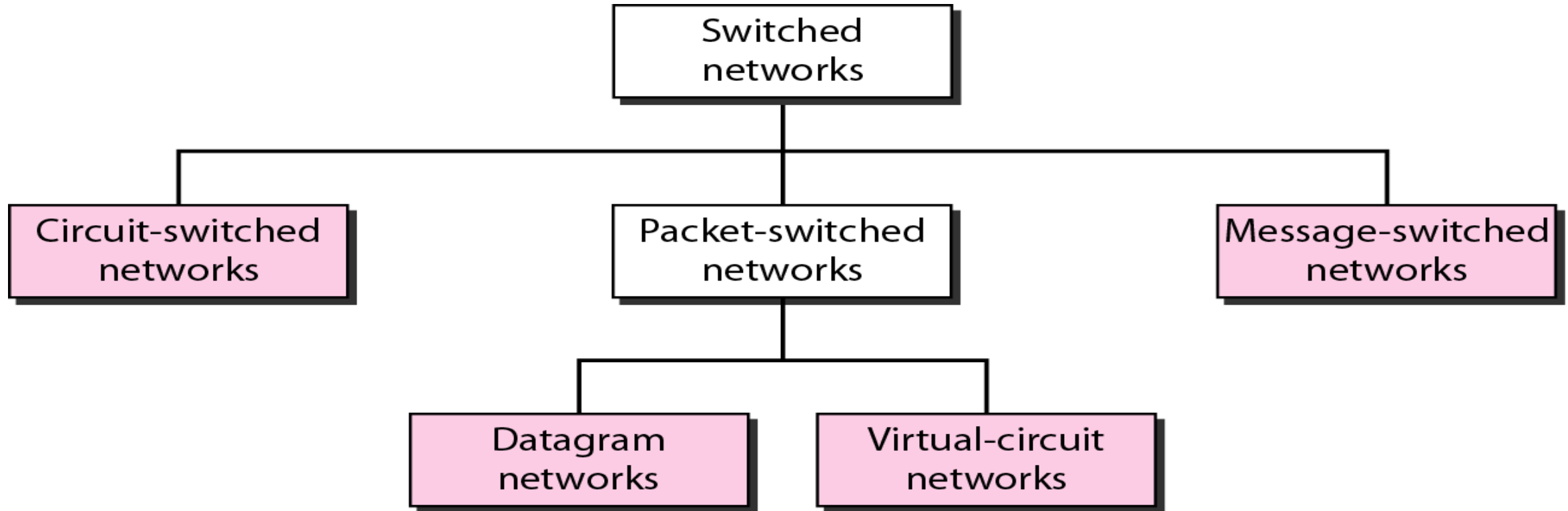
- To add or delete a device requires changing only two connections. The only constraints are media and traffic considerations (maximum ring length and number of devices).
- In addition, fault isolation is simplified. Generally, in a ring a signal is circulating at all times. If one device does not receive a signal within a specified period, it can issue an alarm.
- However, unidirectional traffic can be a disadvantage. In a simple ring, a break in the ring (such as a disabled station) can disable the entire network. This weakness can be solved by using a dual ring or a switch capable of closing off the break.

Hybrid Topology



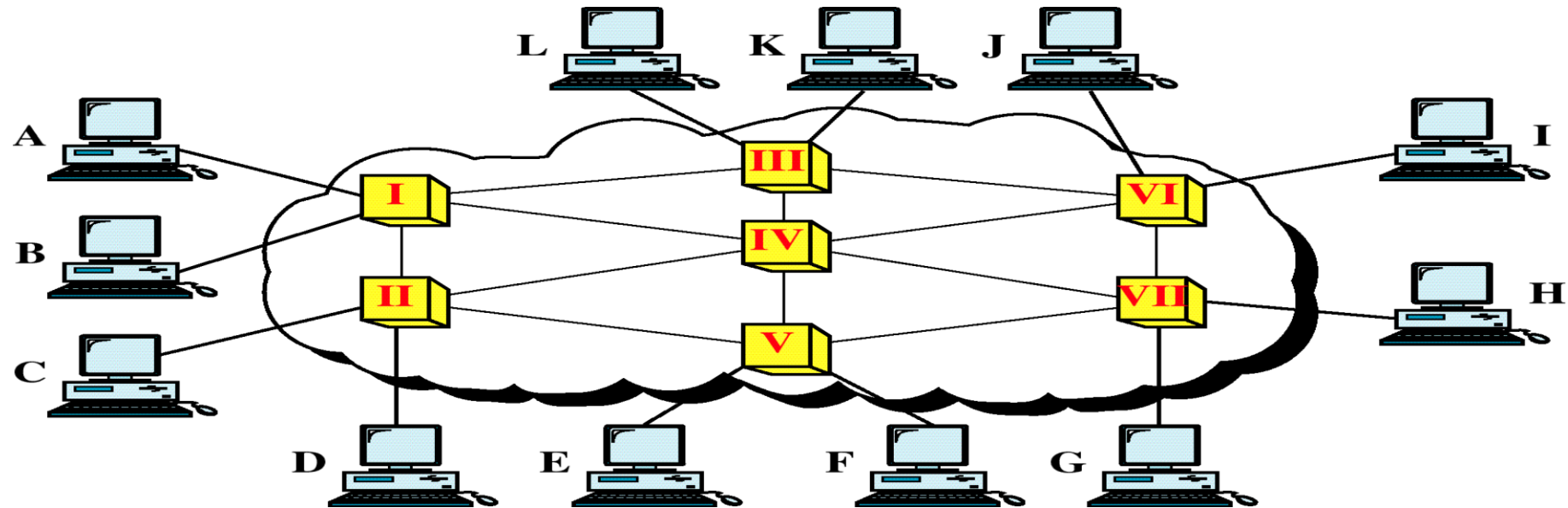
- A hybrid network topology is an interconnection of two or more basic network topologies, each of which contains its own nodes. The resulting topology will exhibit characteristics of all the constituent topologies, thereby limiting the inherent weaknesses of each topology.
- The structure of hybrid topology is more complex but offers various advantages such as flexibility and fault tolerance.

Switching Techniques



Switching

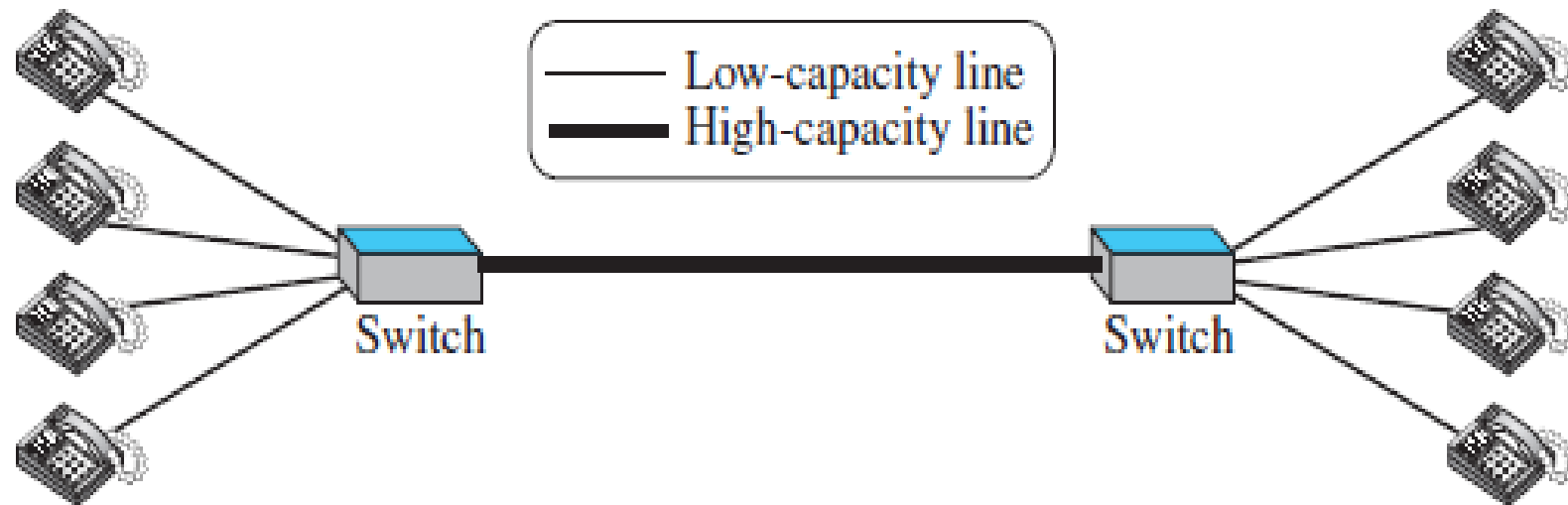
- An internet is a **switched network** in which a switch connects at least two links together. A switch needs to forward data from a network to another network when required.
- The two most common types of switched networks are **circuit-switched** and **packet-switched** networks.



From the above diagram, What are possible ways to send the data from Node C to Node K?

Circuit-Switched Network

- In a **circuit-switched network**, a dedicated connection, called a circuit, is always available between the two end systems; the switch can only make it active or inactive.
- The thick line connecting two switches is a high-capacity communication line that can handle four voice communications at the same time; the capacity can be shared between all pairs of telephone sets.
- Fixed data can be transferred at a time in circuit switching technology.
- The switches used in this example have forwarding tasks but no storing capability.



Circuit-Switched Network

Three Phases

- Connection Setup
 - Transmission Phase
 - Termination Phase
-
- In circuit switching, the resources need to be reserved during the setup phase; the resources remain dedicated for the entire duration of data transfer until the teardown phase.
 - In Transmission phase, actual data communication takes place.
 - Once the transmission is over, the resources has to be released in this phase.
-
- Circuit switching was very common in telephone networks in the past, although part of the telephone network today is a packet-switched network.

Circuit-Switched Network

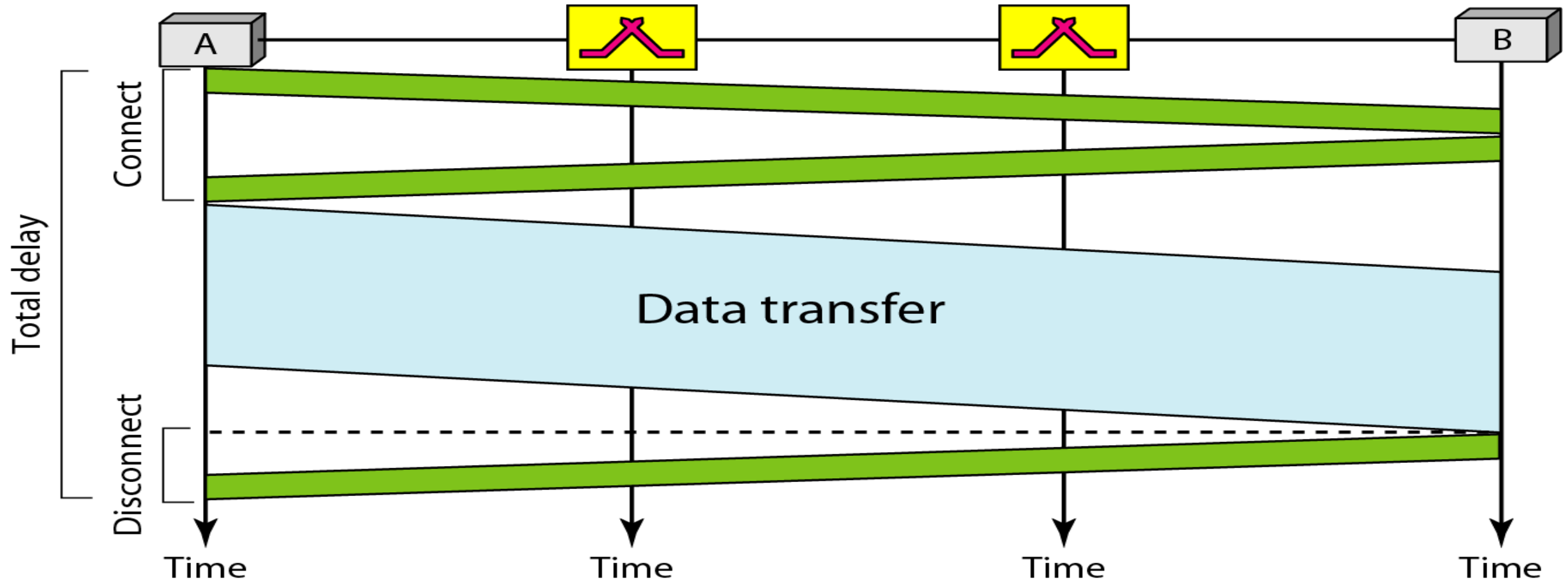
Efficiency

- It can be argued that circuit-switched networks are not as efficient as the other two types of networks because resources are allocated during the entire duration of the connection. These resources are unavailable to other connections. In a telephone network, people normally terminate the communication when they have finished their conversation. However, in computer networks, a computer can be connected to another computer even if there is no activity for a long time. In this case, allowing resources to be dedicated means that other connections are deprived.

Delay:

- Delay Although a circuit-switched network normally has low efficiency, the delay in this type of network is minimal. During data transfer the data are not delayed at each switch; the resources are allocated for the duration of the connection.

Delay in a circuit-switched network



Advantages & Disadvantages

Advantages:

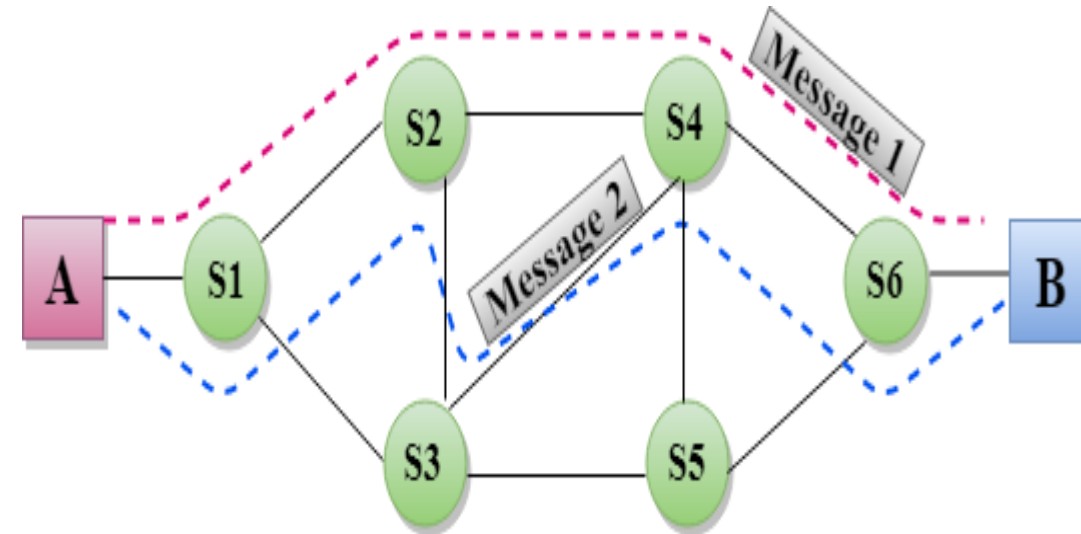
- It has fixed bandwidth since the communication channel is dedicated.

Disadvantages:

- Speed of transmission is fixed after connection establishment.
- It takes a long time to establish a connection approx. 10 seconds during which no data can be transmitted.
- It is more expensive than other switching techniques as a dedicated path is required for each connection.
- It is inefficient to use because once the path is established and no data is transferred, then the capacity of the path is wasted.

Message Switching

- Message Switching is a switching technique in which a message is transferred as a complete unit and routed through intermediate nodes at which it is stored and forwarded.
- In Message Switching technique, there is no establishment of a dedicated path between the sender and receiver.
- The destination address is appended to the message. Message Switching provides a dynamic routing as the message is routed through the intermediate nodes based on the information available in the message.
- Message switches are programmed in such a way so that they can provide the most efficient routes.
- Each and every node stores the entire message and then forward it to the next node. This type of network is known as **store and forward network**.
- Message switching treats each message as an independent entity.



Advantages & Disadvantages of Message Switching

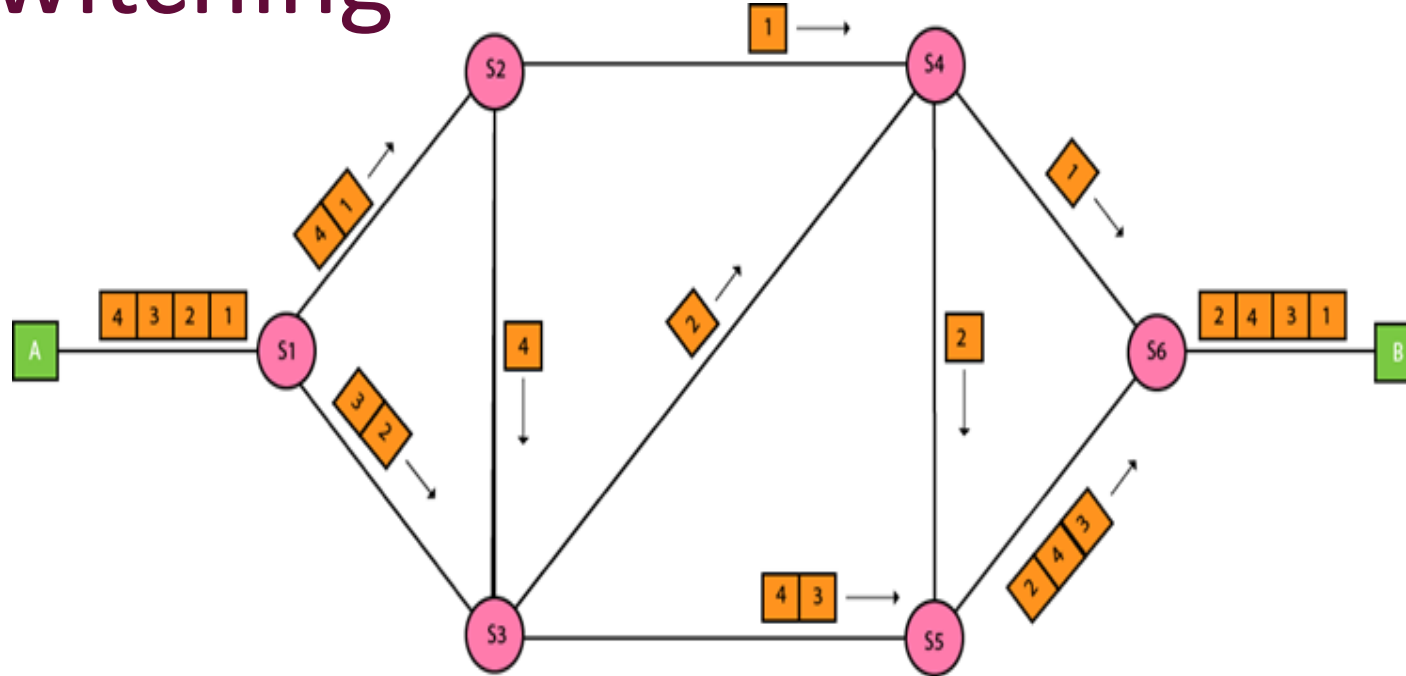
Advantages:

- Data channels are shared among the communicating devices that improve the efficiency of using available bandwidth.
- Traffic congestion can be reduced because the message is temporarily stored in the nodes.
- Message priority can be used to manage the network.
- The size of the message which is sent over the network can be varied. Therefore, it supports the data of unlimited size.

Disadvantages:

- The message switches must be equipped with sufficient storage to enable them to store the messages until the message is forwarded.
- The Long delay can occur due to the storing and forwarding facility provided by the message switching technique.

Packet Switching



- The message is divided into smaller pieces called packets and they are sent individually.
- packets are given a unique number to identify their order at the receiving end.
- Every packet contains some information in its headers such as source address, destination address and sequence number.
- All the packets are reassembled at the receiving end in correct order.
- If any packet is missing or corrupted, then the message will be sent to resend the message.
- If the correct order of the packets is reached, then the acknowledgment message will be sent.

Approaches of Packet Switching

There are two approaches to Packet Switching:

Datagram Packet switching:

- It is a packet switching technology in which packet is known as a datagram, is considered as an independent entity. Each packet contains the information about the destination and switch uses this information to forward the packet to the correct destination.
- The packets are reassembled at the receiving end in correct order.
- In Datagram Packet Switching technique, the path is not fixed.
- Intermediate nodes take the routing decisions to forward the packets.
- Datagram Packet Switching is also known as connectionless switching.

Virtual Circuit Switching:

- Virtual Circuit Switching is also known as connection-oriented switching.
- In the case of Virtual circuit switching, a preplanned route is established before the messages are sent.
- Call request and call accept packets are used to establish the connection between sender and receiver.
- In this case, the path is fixed for the duration of a logical connection.

Advantages of Packet Switching

- **Cost-effective:** In packet switching technique, switching devices do not require massive secondary storage to store the packets, so cost is minimized to some extent. Therefore, we can say that the packet switching technique is a cost-effective technique.
- **Reliable:** If any node is busy, then the packets can be rerouted. This ensures that the Packet Switching technique provides reliable communication.
- **Efficient:** Packet Switching is an efficient technique. It does not require any established path prior to the transmission, and many users can use the same communication channel simultaneously, hence makes use of available bandwidth very efficiently.

Disadvantages of Packet Switching

- Packet Switching technique cannot be implemented in those applications that require low delay and high-quality services.
- The protocols used in a packet switching technique are very complex and requires high implementation cost.
- If the network is overloaded or corrupted, then it requires retransmission of lost packets. It can also lead to the loss of critical information if errors are not recovered.

*We use the concept of **layers** in our daily life. As an example, let us consider two friends who communicate through postal mail. The process of sending a letter to a friend would be complex if there were no services available from the post office.*

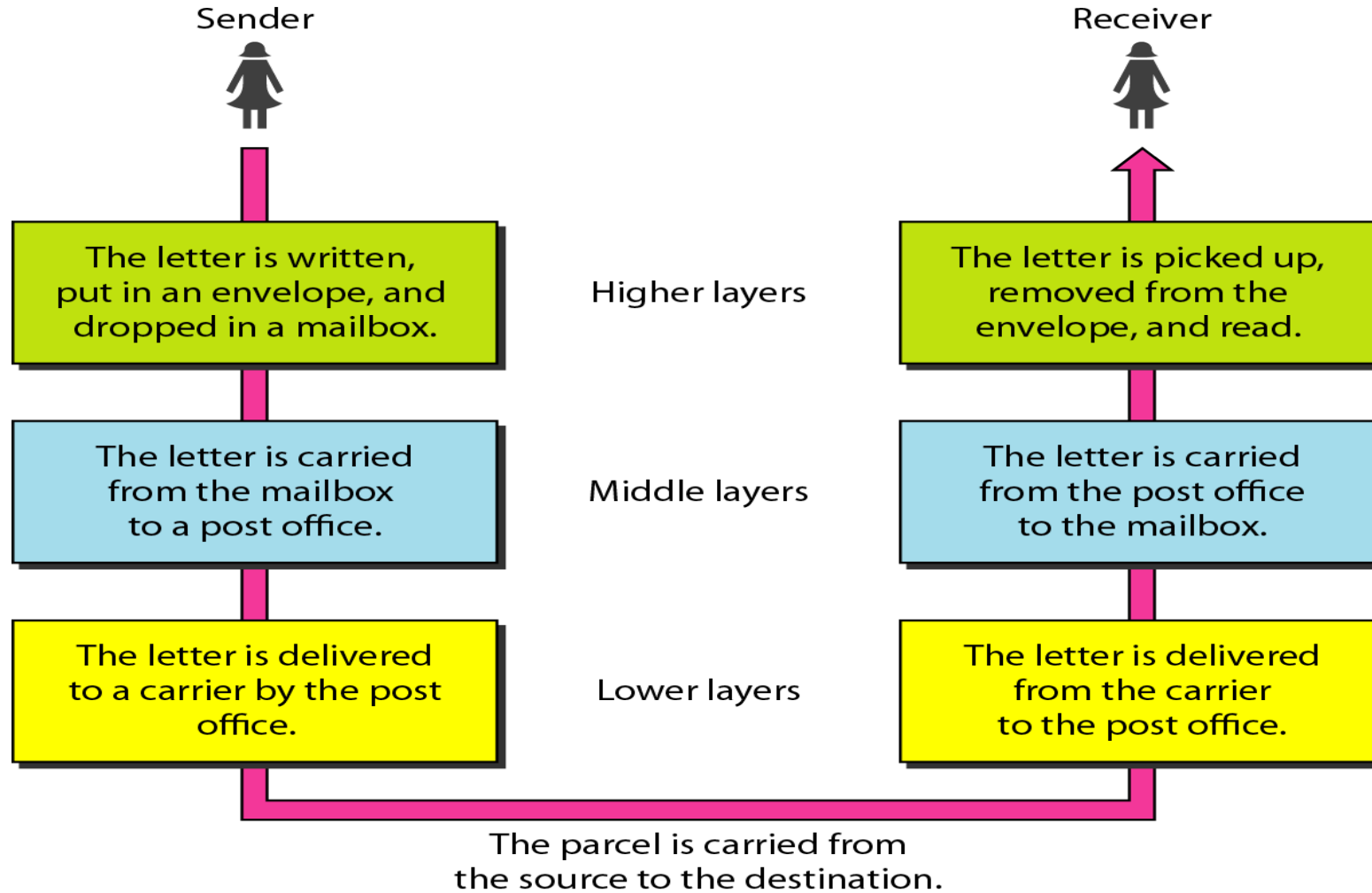
Topics discussed in this section:

Sender, Receiver, and Carrier
Hierarchy

Layer Architecture

- ❏ Layer architecture simplifies the network design.
- ❏ It is easy to debug network applications in a layered architecture network.
- ❏ The network management is easier due to the layered architecture.
- ❏ Network layers follow a set of rules, called protocol.
- ❏ The protocol defines the format of the data being exchanged, and the control and timing for the handshake between layers.

Tasks involved in sending a letter



THE OSI MODEL

*Established in 1947, the International Standards Organization (**ISO**) is a multinational body dedicated to worldwide agreement on international standards. An ISO standard that covers all aspects of network communications is the Open Systems Interconnection (**OSI**) model. It was first introduced in the late 1970s.*

Topics discussed in this section:

- Layered Architecture
- Peer-to-Peer Processes
- Encapsulation

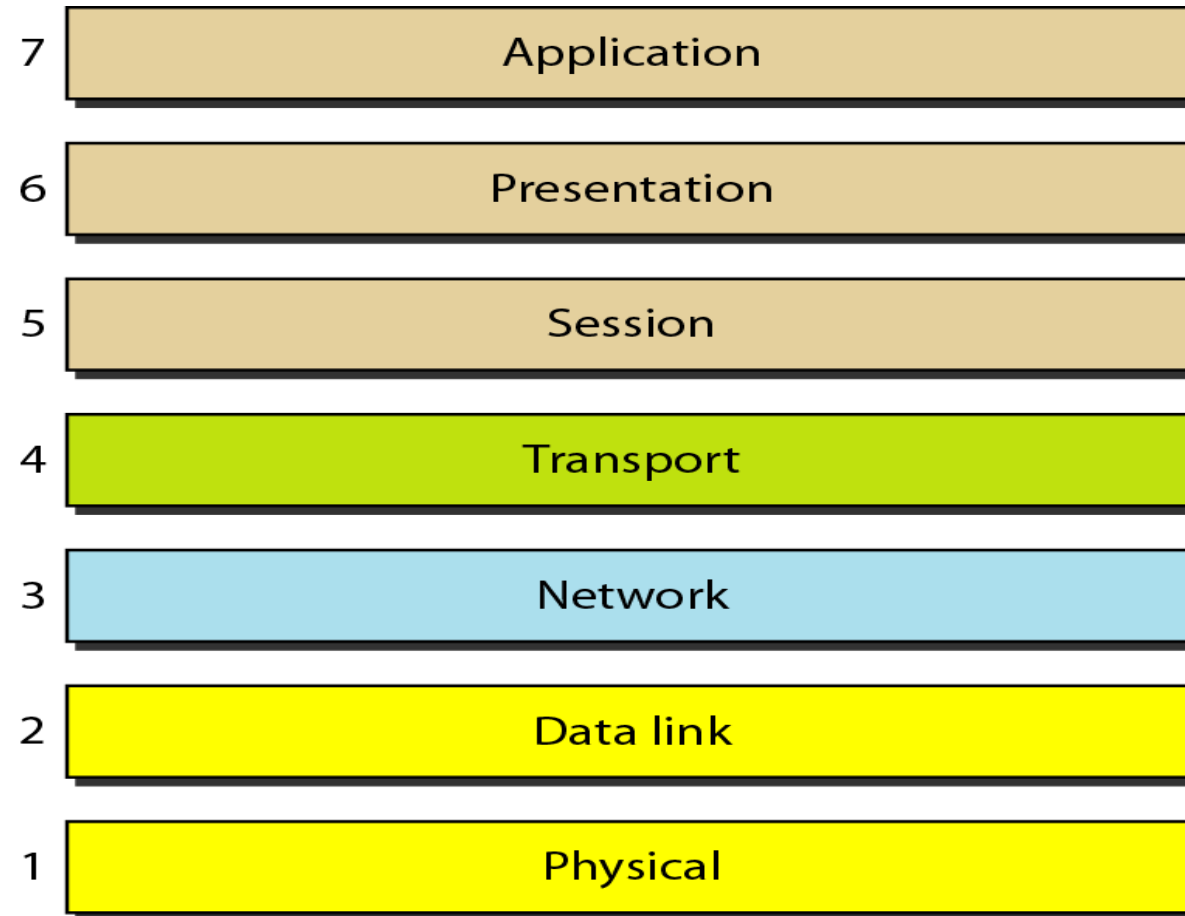
Open Systems Interconnection (OSI) Model

- ❑ International standard organization (ISO) established a committee in 1977 to develop an architecture for computer communication.
- ❑ Open Systems Interconnection (OSI) reference model is the result of this effort.
- ❑ In 1984, the Open Systems Interconnection (OSI) reference model was approved as an international standard for communications architecture.
- ❑ Term “open” denotes the ability to connect any two systems which conform to the reference model and associated standards.

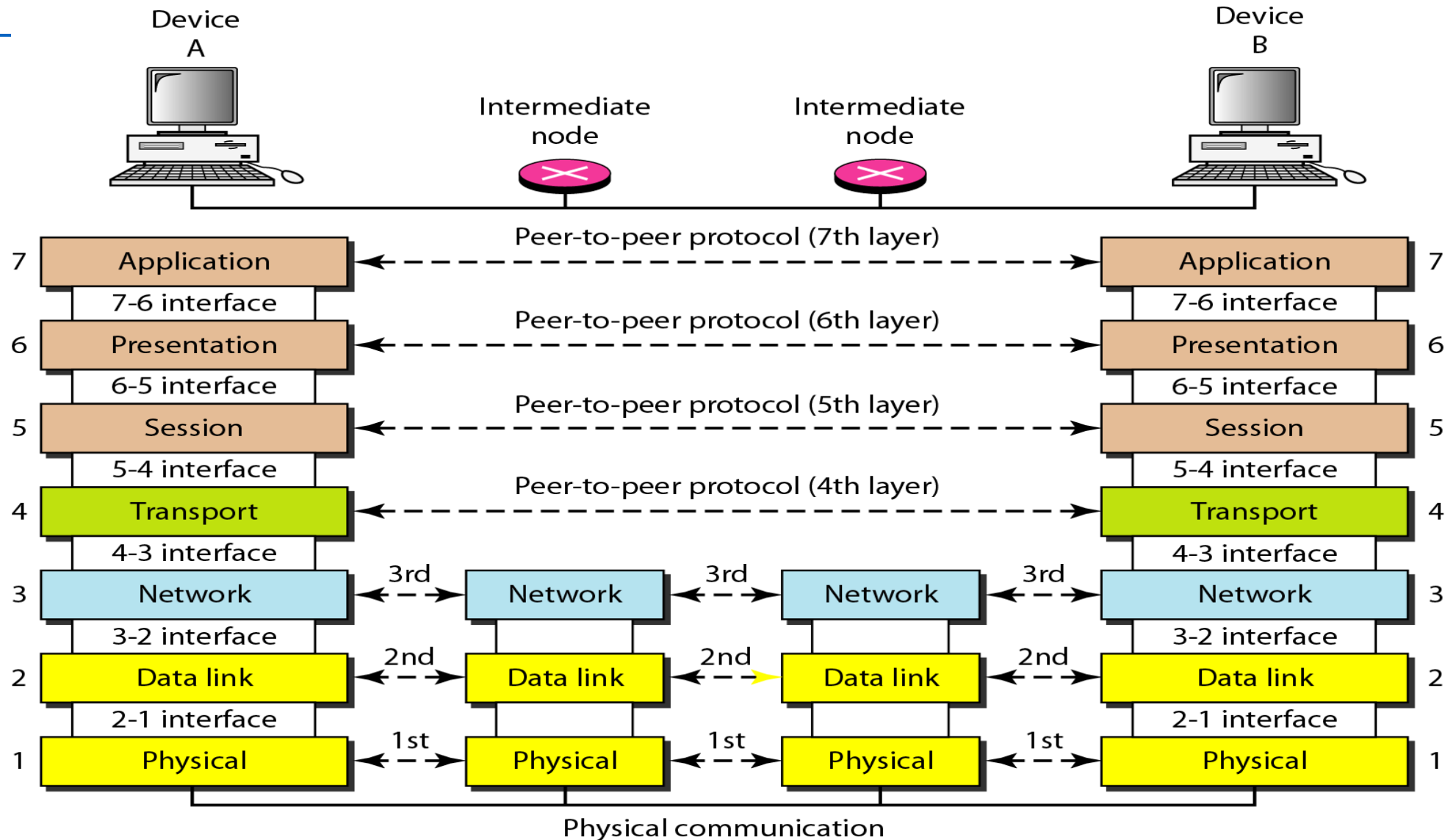
OSI Reference Model

-  The OSI model is now considered the primary Architectural model for inter-computer communications.
-  The OSI model describes how information or data makes its way from application programmes (such as spreadsheets) through a network medium (such as wire) to another application programme located on another network.
-  The OSI reference model divides the problem of moving information between computers over a network medium into SEVEN smaller and more manageable problems .
-  This separation into smaller more manageable functions is known as layering.

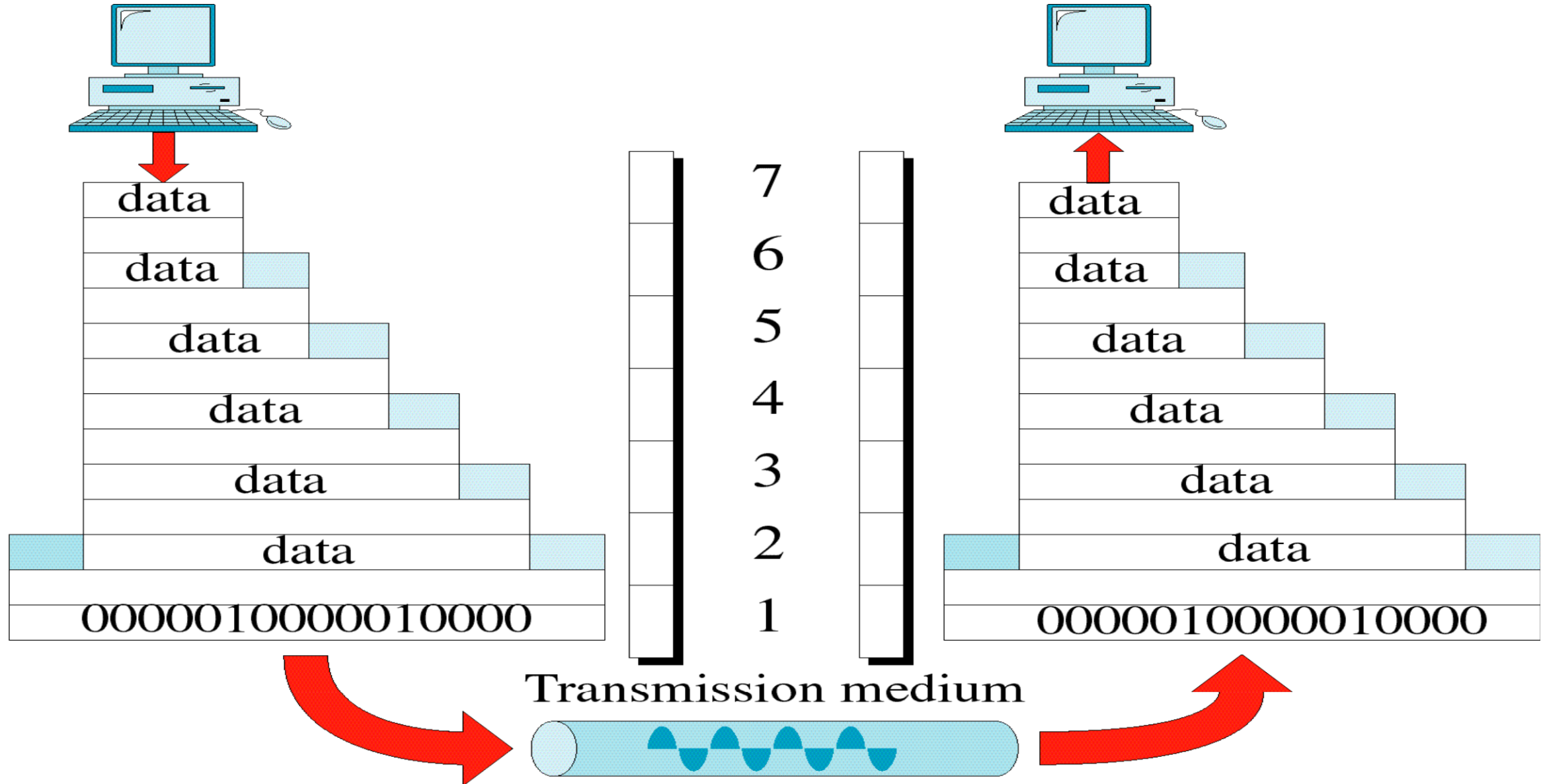
Seven layers of the OSI model



The interaction between layers in the OSI model

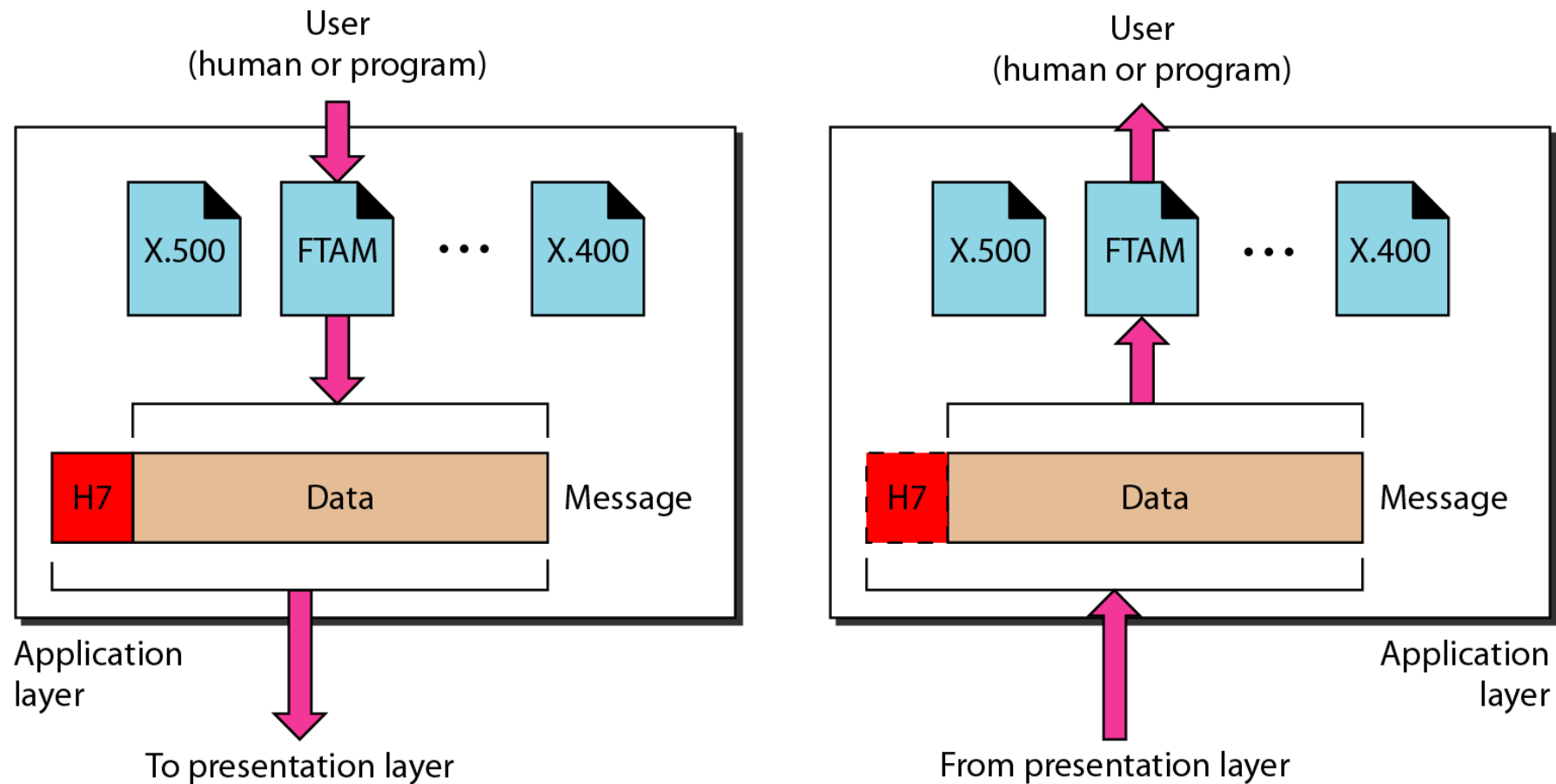


An Exchange Using the OSI Model



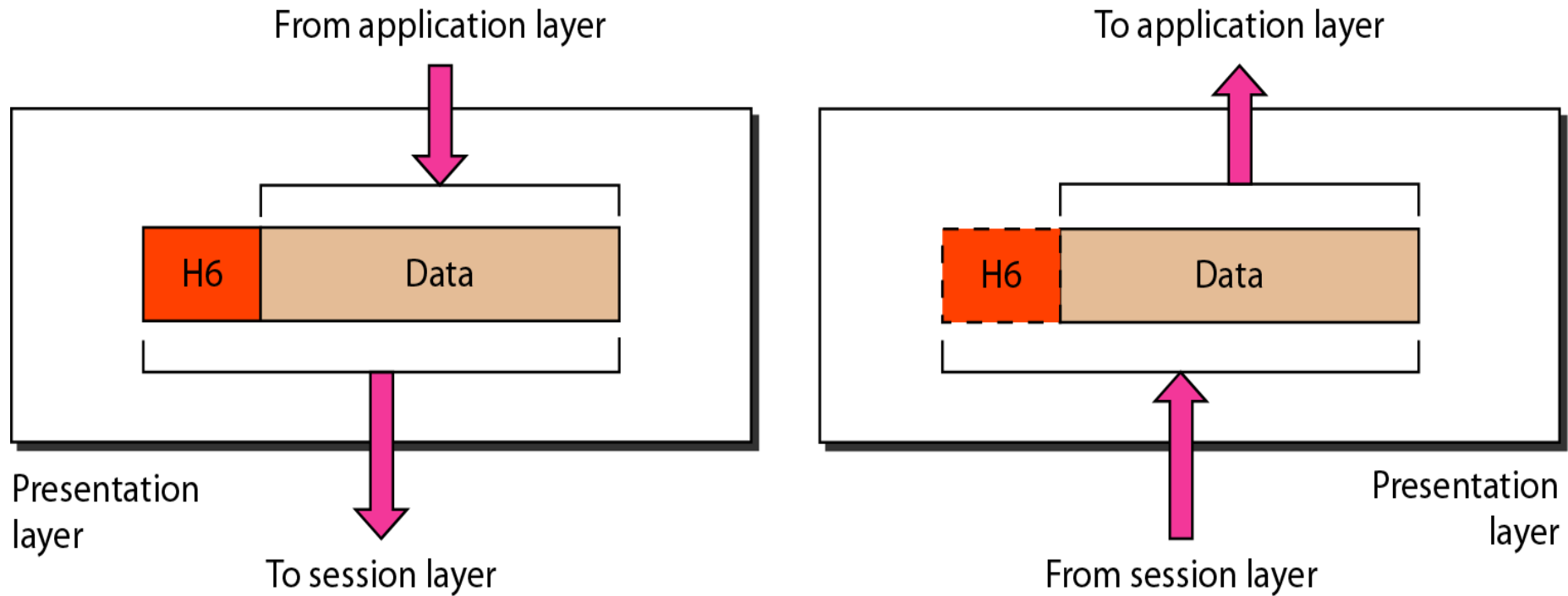
Application Layer

The application layer is responsible for providing services to the user.



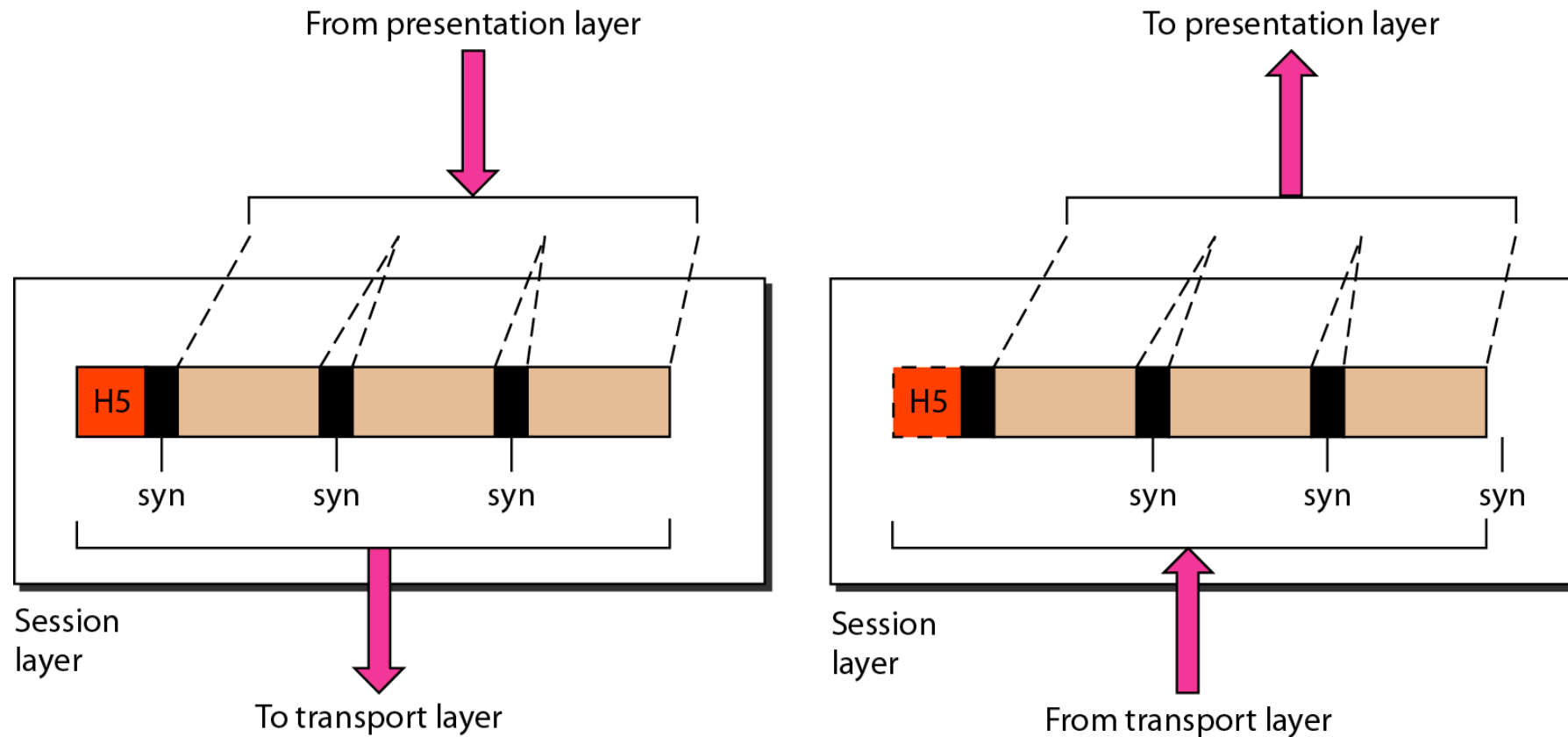
Presentation Layer

The presentation layer is responsible for translation, compression, and encryption.



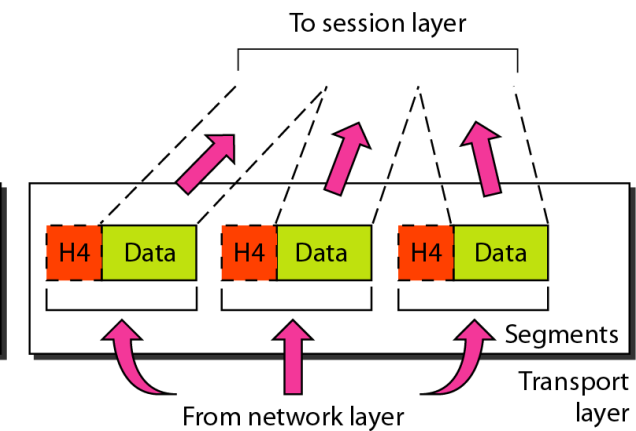
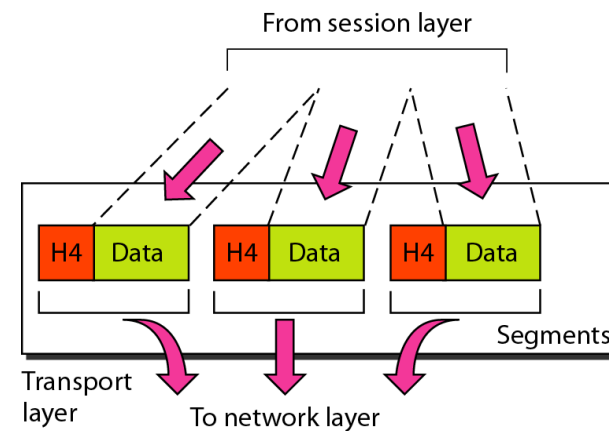
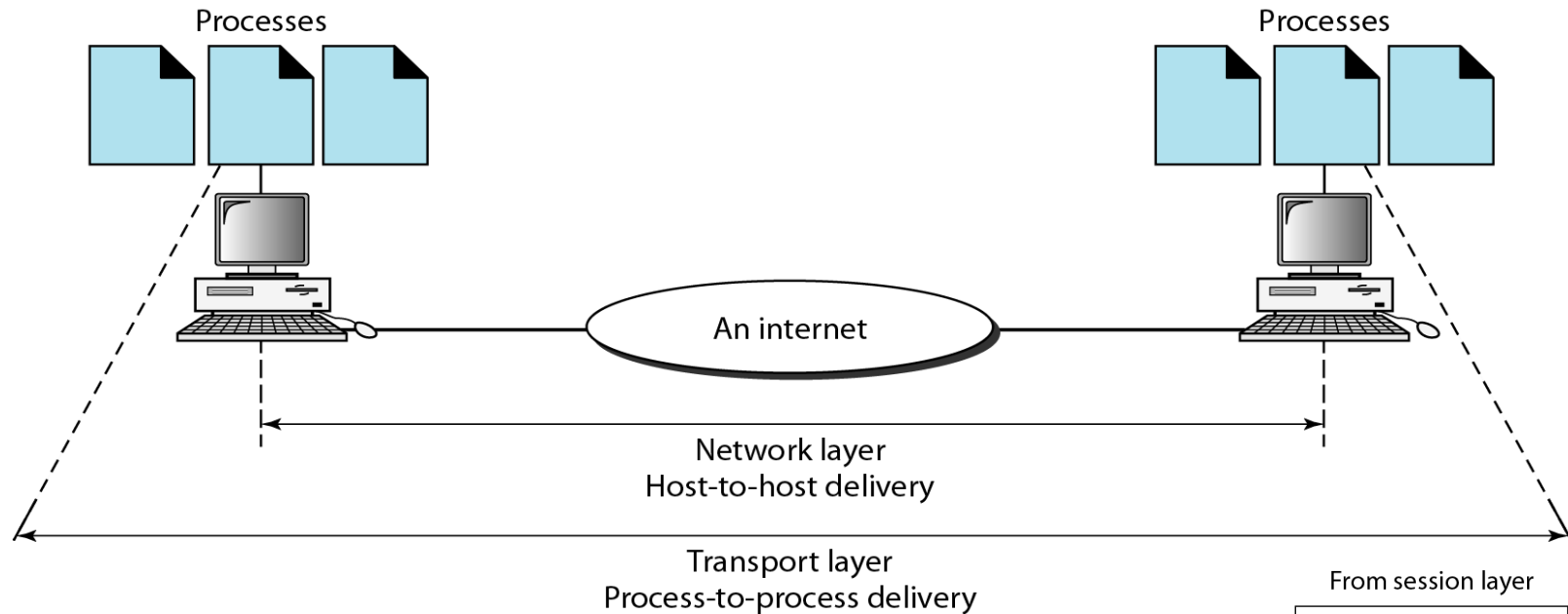
Session Layer

The session layer is responsible for dialog control and synchronization.



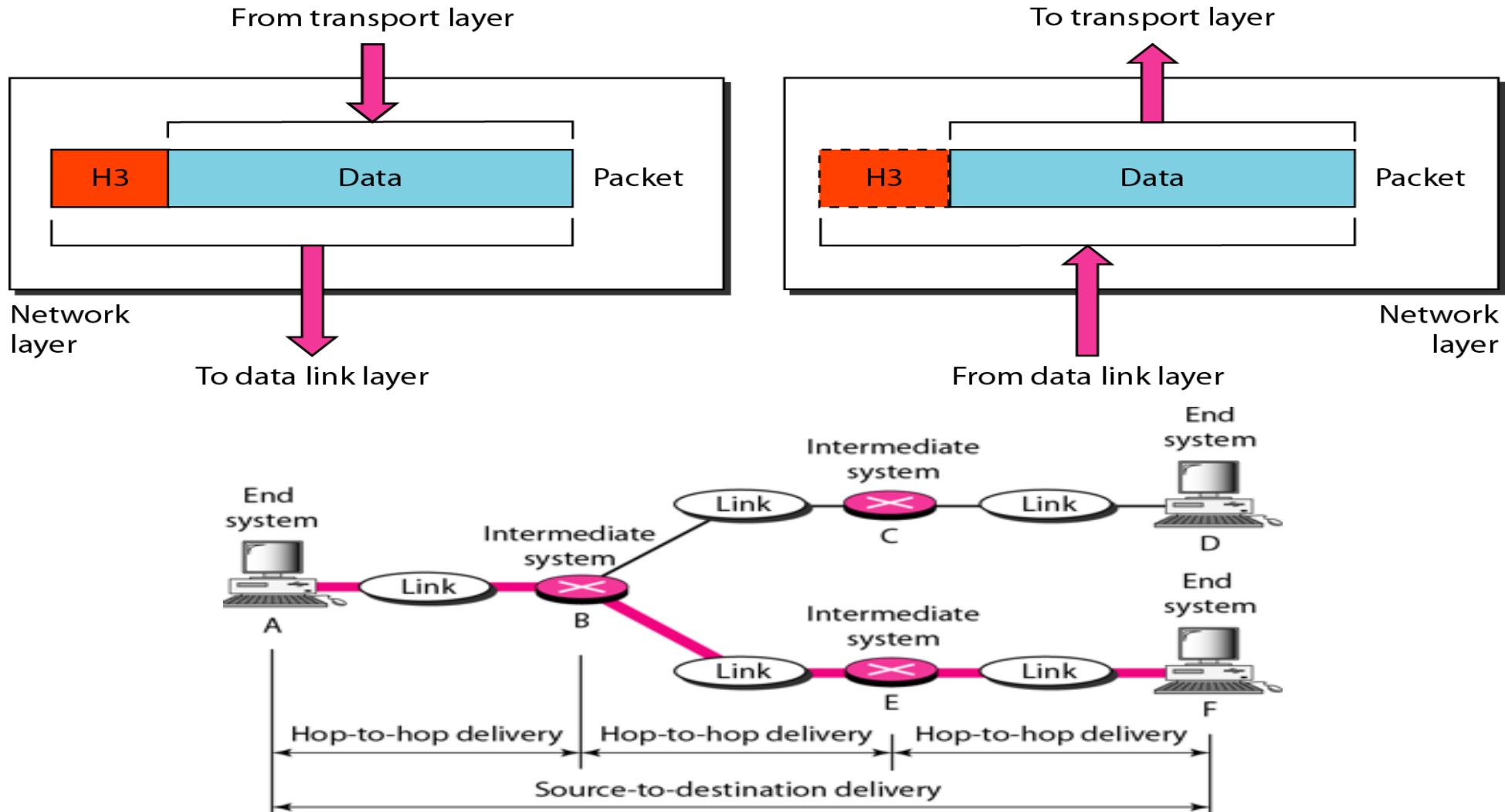
Transport Layer

The transport layer is responsible for the delivery of a message from one process to another.



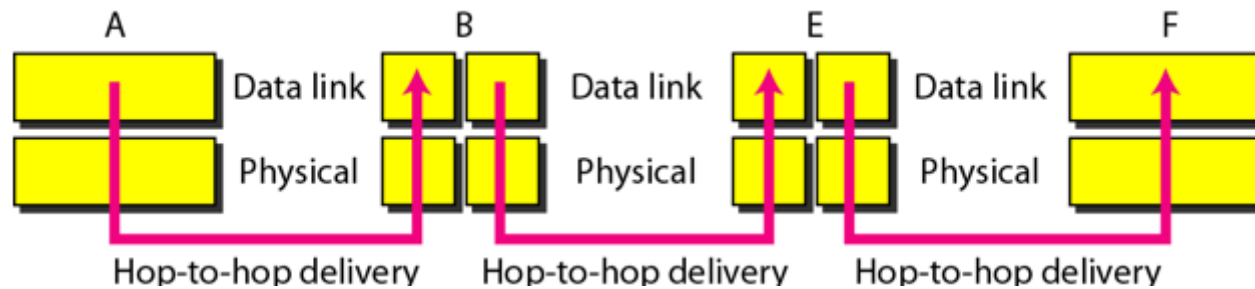
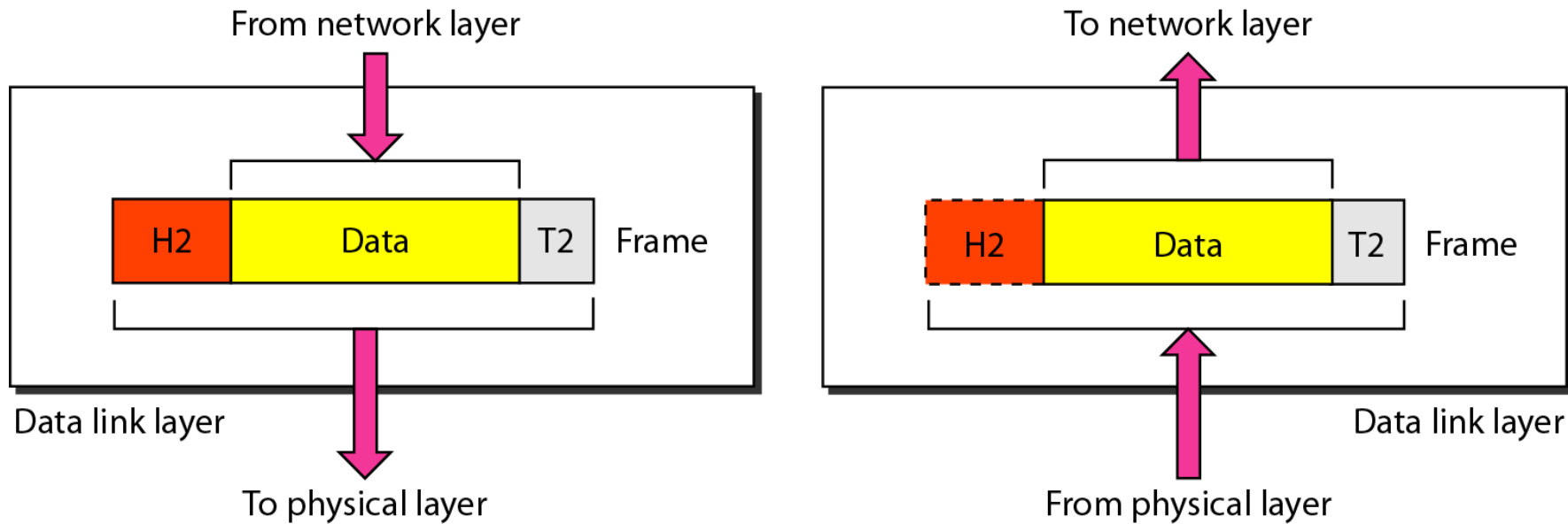
Network Layer

The network layer is responsible for the delivery of individual packets from the source host to the destination host.



Data Link Layer

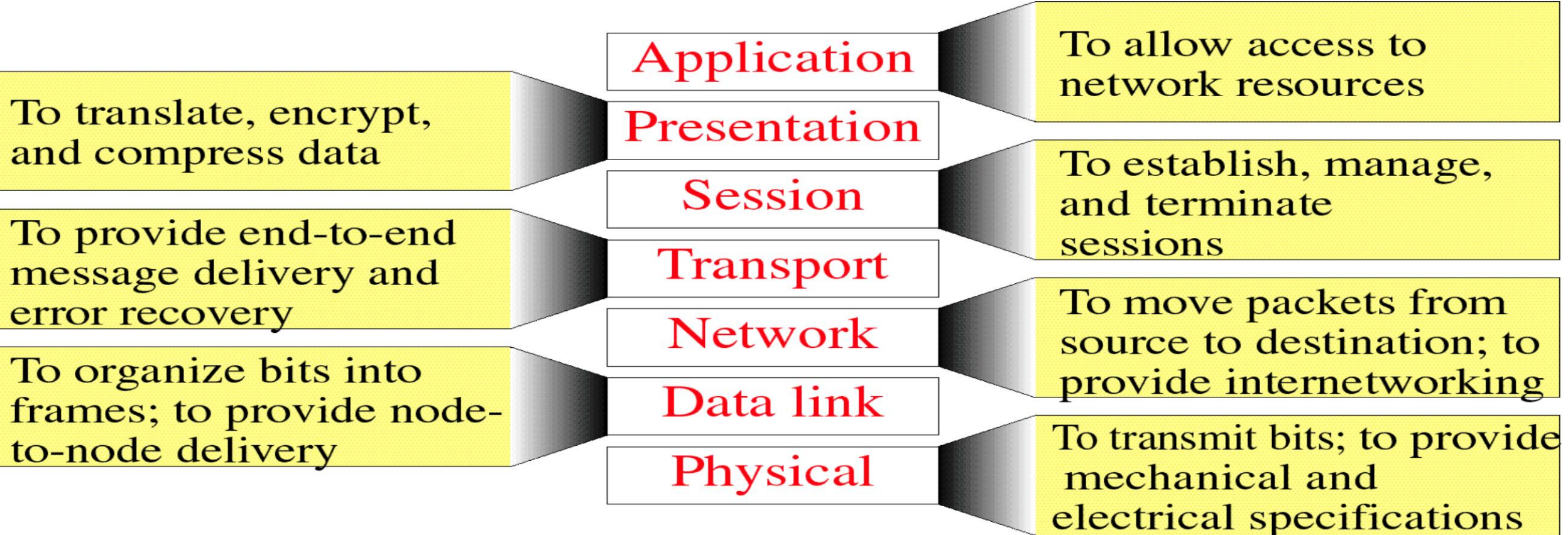
The data link layer is responsible for moving frames from one hop (node) to the next.



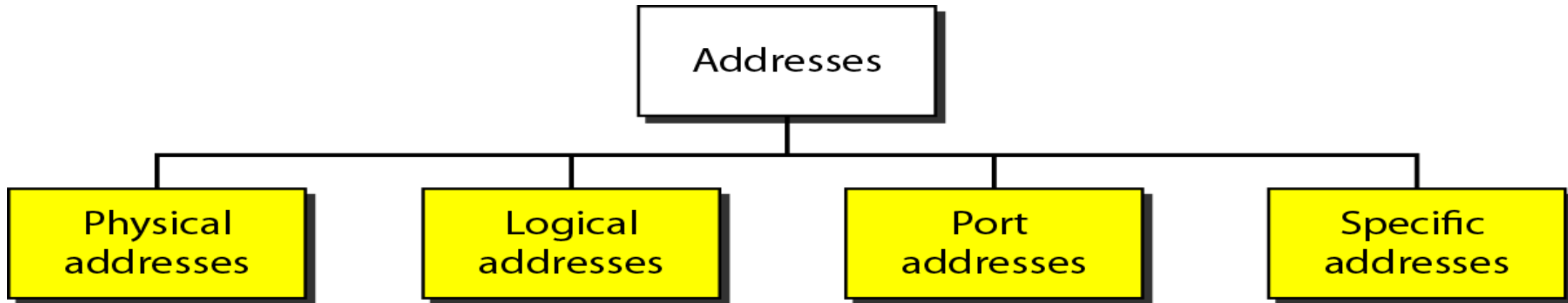
Physical Layer

- ❏ Provides physical interface for transmission of information.
- ❏ Defines rules by which bits are passed from one system to another on a physical communication medium.
- ❏ Covers all - mechanical, electrical, functional and procedural - aspects for physical communication.
- ❏ Such characteristics as voltage levels, timing of voltage changes, physical data rates, maximum transmission distances, physical connectors, and other similar attributes are defined by physical layer specifications.

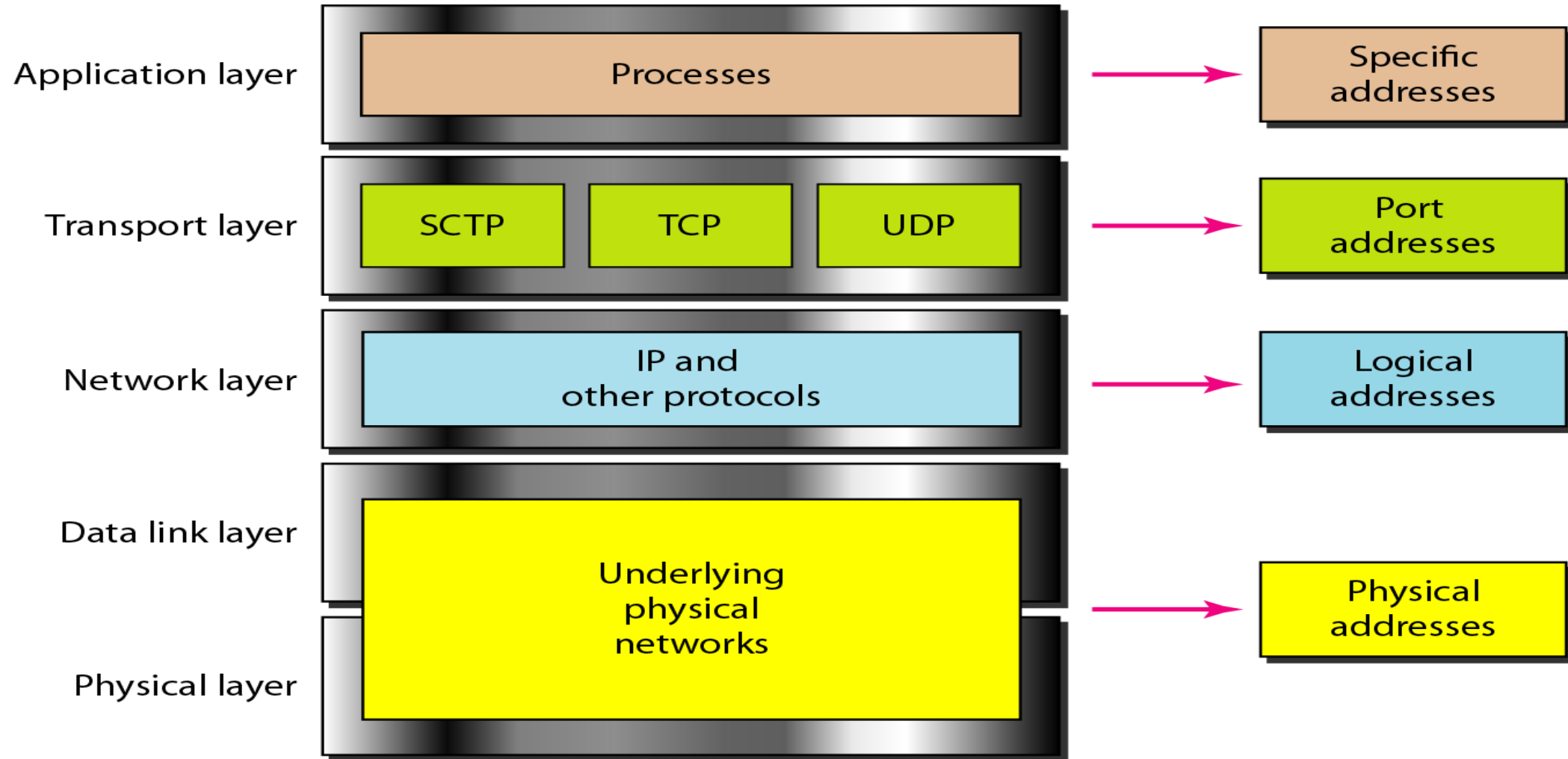
Summary of Layer Functions



The layers in the *TCP/IP protocol suite* do not exactly match those in the OSI model. The original TCP/IP protocol suite was defined as having four layers: *host-to-network*, *internet*, *transport*, and *application*. However, when TCP/IP is compared to OSI, we can say that the TCP/IP protocol suite is made of five layers: *physical*, *data link*, *network*, *transport*, and *application*.



Relationship of layers and addresses in TCP/IP



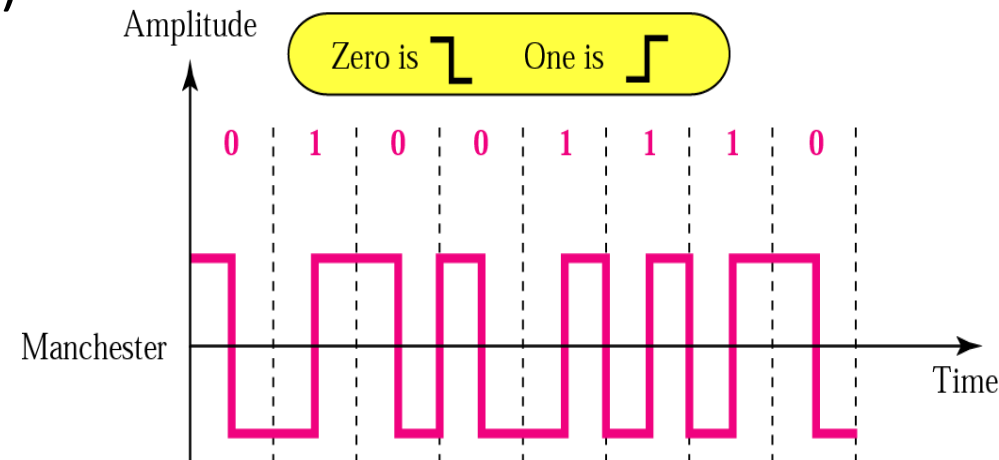
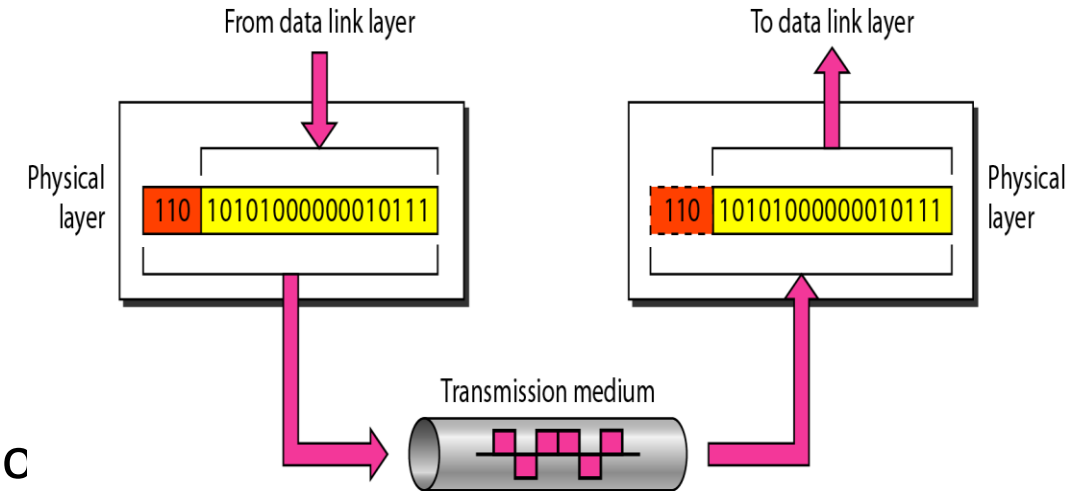
Physical Layer

• Functions

- Transmission of a raw bit stream
- Forms the physical interface between devices

• Issues

- Which modulation technique (bits to pulse (analog signal))?
- Which Line Coding technique (bits to digital signal?)
- How long will a bit last? (*bit interval vs bit rate*)
- Bit- serial or parallel transmission?
- Half- or Full- duplex transmission?
- How many pins does the network connector have?
- How is a connection set up or torn down?



Performance

One important issue in networking is the **performance** of the network—how good is it? We discuss quality of service, an overall measurement of network performance.

- Bandwidth - capacity of the system
- Throughput - no. of bits that can be pushed through.
- Latency (Delay) - delay incurred by a bit from start to finish
- Bandwidth-Delay Product

Note: In networking, we use the term bandwidth in two contexts.

- The first, bandwidth in **hertz**, refers to the range of frequencies in a composite signal or the range of frequencies that a channel can pass.
- The second, bandwidth in **bits per second**, refers to the speed of bit transmission in a channel or link. Often referred to as Capacity.

Example

A network with bandwidth of 10 Mbps can pass only an average of 12,000 frames per minute with each frame carrying an average of 10,000 bits. What is the throughput of this network?

Solution

We can calculate the throughput as

$$\text{Throughput} = \frac{12,000 \times 10,000}{60} = 2 \text{ Mbps}$$

The throughput is almost one-fifth of the bandwidth in this case.

Propagation & Transmission delay

- Propagation speed - speed at which a bit travels through the medium from source to destination.
- Transmission speed - the speed at which all the bits in a message arrive at the destination. (difference in arrival time of first and last bit)
 - **Propagation Delay** = Distance/Propagation speed
 - **Transmission Delay** = Message size/bandwidth bps
 - **Latency** = Propagation delay + Transmission delay + Queueing time + Processing time

Example

What are the propagation time and the transmission time for a 2.5-kbyte message (an e-mail) if the bandwidth of the network is 1 Gbps? Assume that the distance between the sender and the receiver is 12,000 km and that light travels at 2.4×10^8 m/s.

Solution

- **Propagation Delay** = Distance/Propagation speed

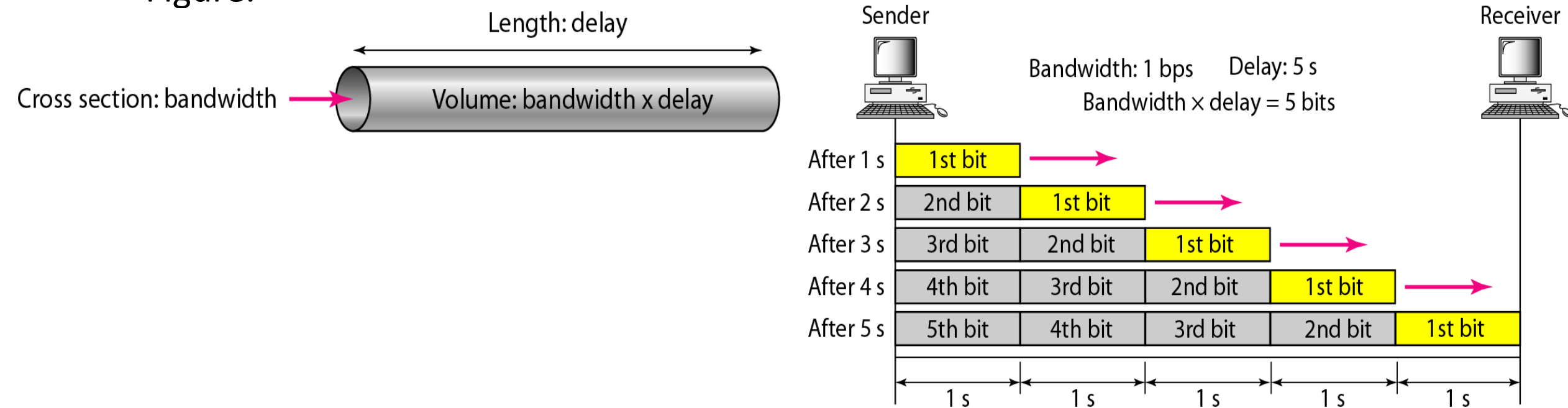
$$\text{Propagation time} = \frac{12,000 \times 1000}{2.4 \times 10^8} = 50 \text{ ms}$$

$$\text{Transmission time} = \frac{2500 \times 8}{10^9} = 0.020 \text{ ms}$$

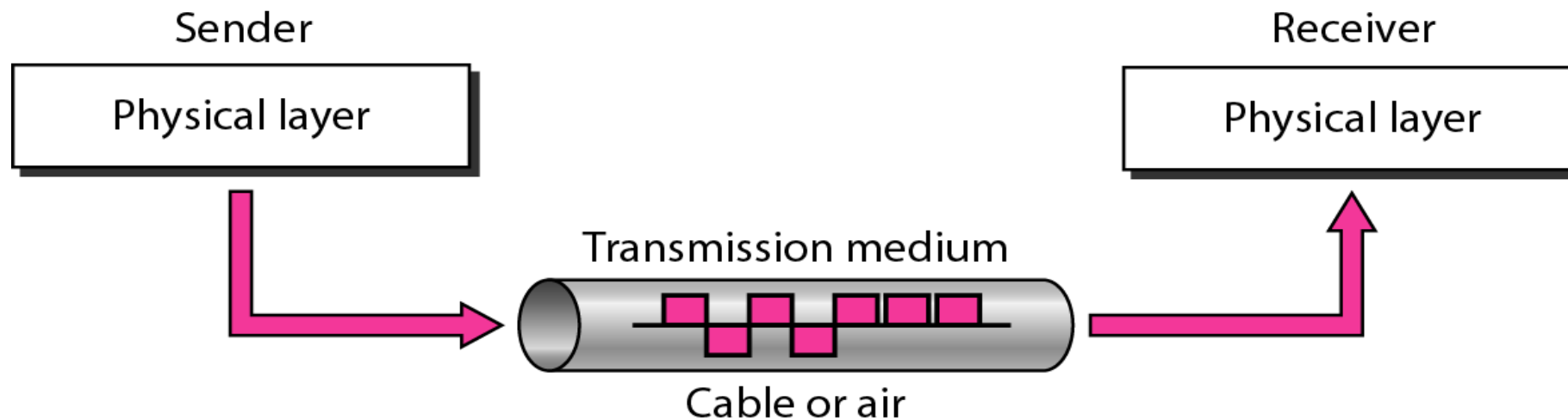
Bandwidth-Delay Product

The bandwidth-delay product defines the number of bits that can fill the link.

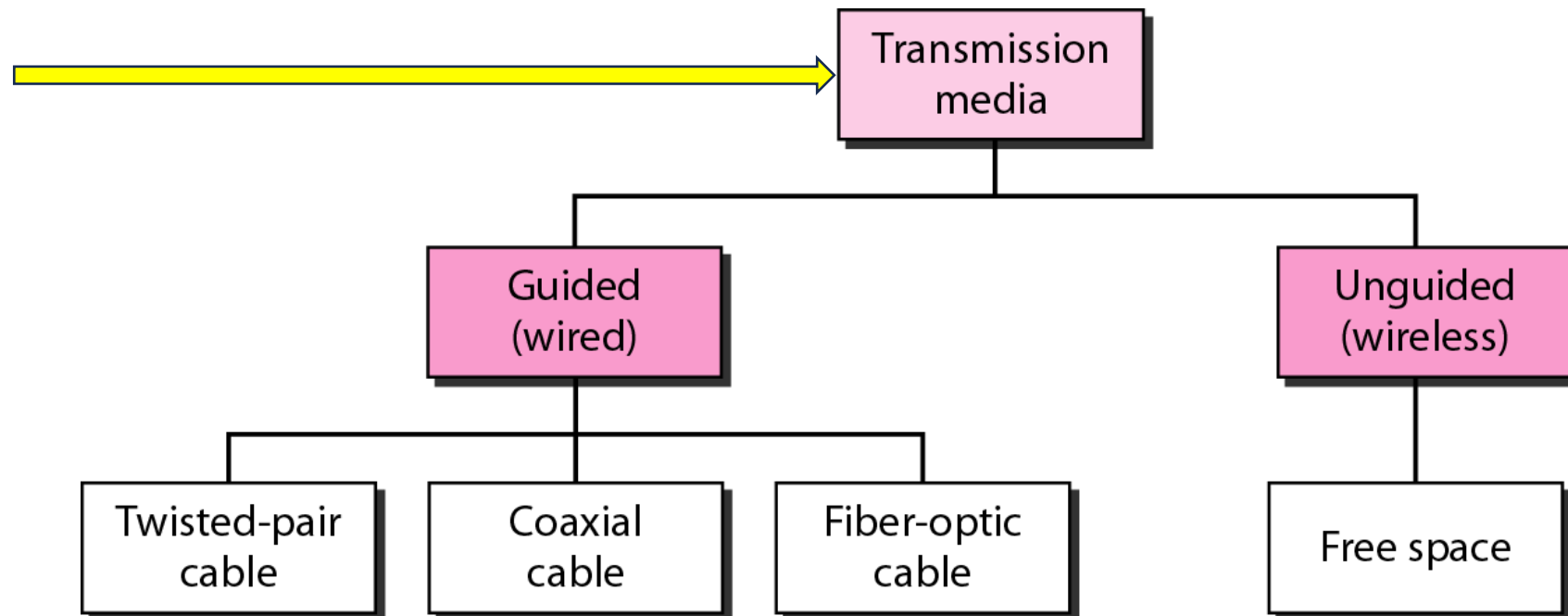
- We can think about the link between two points as a pipe. The cross section of the pipe represents the bandwidth, and the length of the pipe represents the delay. We can say the volume of the pipe defines the bandwidth-delay product, as shown in Figure.



Transmission Media



TYPES



Guided Media

- *Transmission media is a communication channel that carries the information from the sender to the receiver. Data is transmitted through the electromagnetic signals.*
- *Guided media, which are those that provide a conduit from one device to another, include twisted-pair cable, coaxial cable, and fiber-optic cable.*
- *It is defined as the physical medium through which the signals are transmitted.*

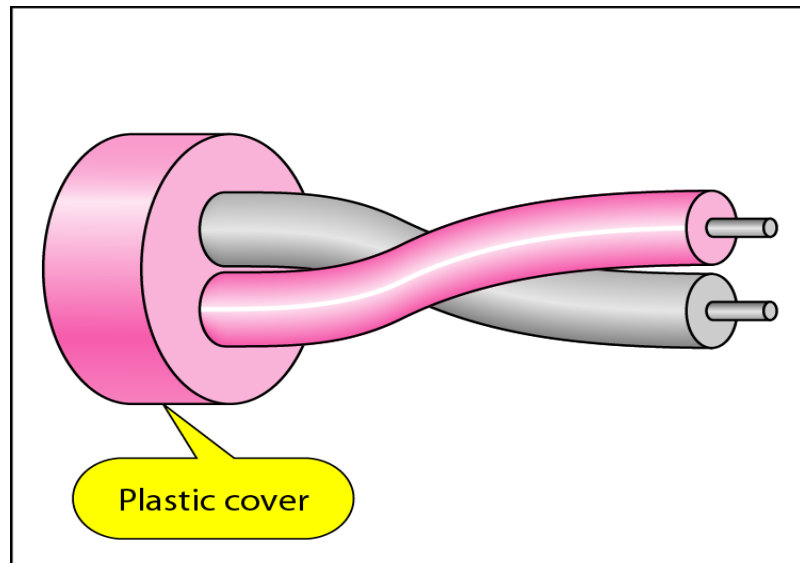
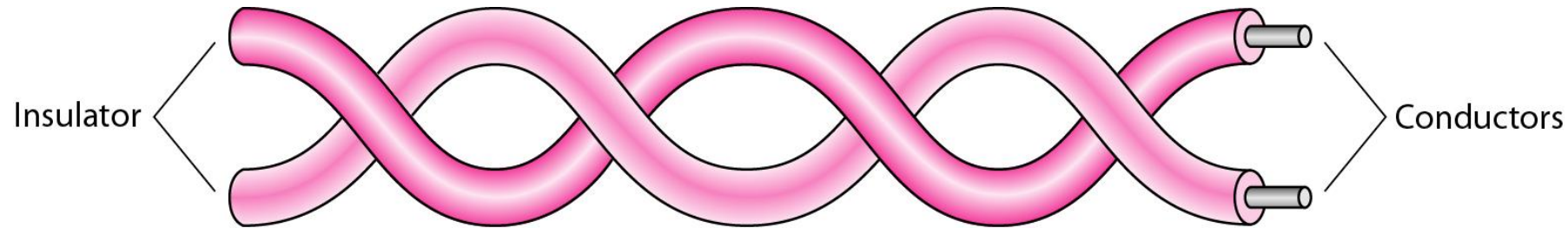
Types

- Twisted-Pair Cable
- Coaxial Cable
- Fiber-Optic Cable

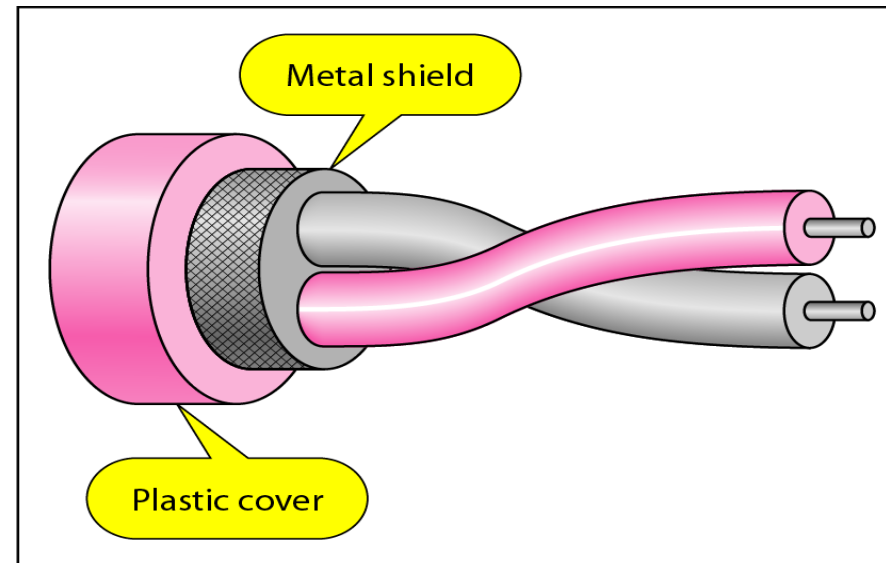
Twisted-Pair Cable

Twisted pair is a physical media made up of a pair of cables twisted with each other. A twisted pair cable is cheap as compared to other transmission media. Installation of the twisted pair cable is easy, and it is a lightweight cable. The frequency range for twisted pair cable is from 0 to 3.5KHz.

The degree of reduction in noise interference is determined by the number of turns per foot. Increasing the number of turns per foot decreases noise interference.



a. UTP



b. STP

UTP Vs. STP

An unshielded twisted pair is widely used in telecommunication. Following are the categories of the UTP cable:

- **Category 1:** Category 1 is used for telephone lines that have low-speed data.
- **Category 2:** It can support upto 4Mbps.
- **Category 3:** It can support upto 16Mbps.
- **Category 4:** It can support upto 20Mbps.
Therefore, it can be used for long distance communication.
- **Category 5:** It can support upto 200Mbps.

Advantages Of UTP:

- It is cheap.
- Installation of the unshielded twisted pair is easy.
- It can be used for high-speed LAN.

Disadvantage:

- This cable can only be used for shorter distances because of attenuation.

A shielded twisted pair is a cable that contains the mesh surrounding the wire that allows the higher transmission rate.

Characteristics of STP:

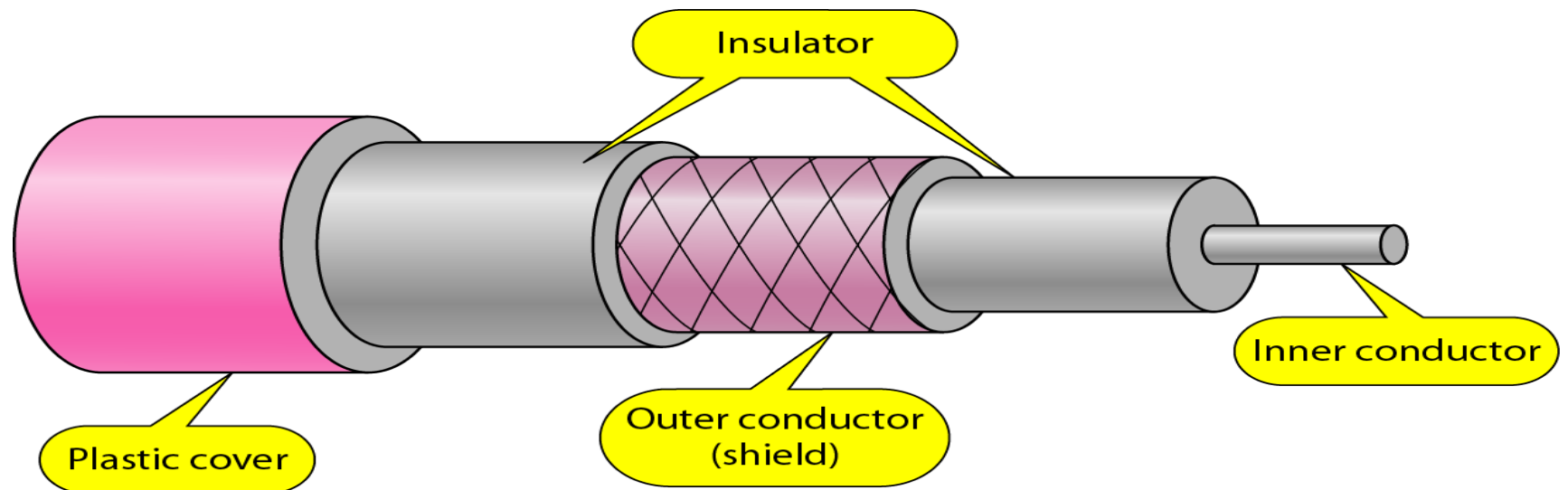
- The cost of the shielded twisted pair cable is not very high and not very low.
- An installation of STP is easy.
- It has higher capacity as compared to unshielded twisted pair cable.
- It has a higher attenuation.
- It is shielded that provides the higher data transmission rate.

Disadvantages

- It is more expensive as compared to UTP and coaxial cable.
- It has a higher attenuation rate.

Coaxial Cable

- Coaxial cable is very commonly used transmission media, for example, TV wire is usually a coaxial cable.
- The name of the cable is coaxial as it contains two conductors parallel to each other.
- It has a higher frequency as compared to Twisted pair cable.
- The inner conductor of the coaxial cable is made up of copper, and the outer conductor is made up of copper mesh.
- The middle core is made up of nonconductive cover that separates the inner conductor from the outer conductor.
- The middle core is responsible for the data transferring whereas the copper mesh prevents from the EMI(Electromagnetic interference).



Coaxial Cable

Coaxial cable is of two types:

1. **Baseband transmission:**

- It is defined as the process of transmitting a single signal at high speed.

2. **Broadband transmission:**

- It is defined as the process of transmitting multiple signals simultaneously.

Advantages of Coaxial cable:

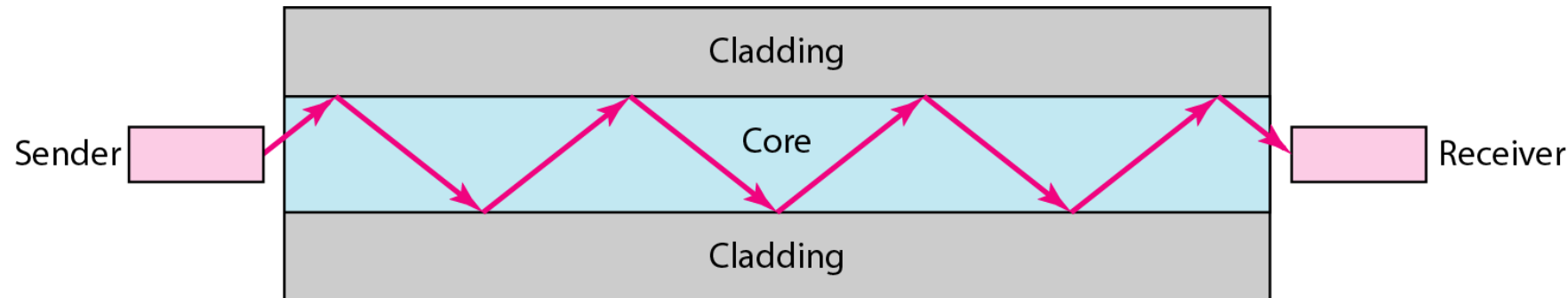
- The data can be transmitted at high speed.
- It has better shielding as compared to twisted pair cable.
- It provides higher bandwidth.

Disadvantages Of Coaxial cable:

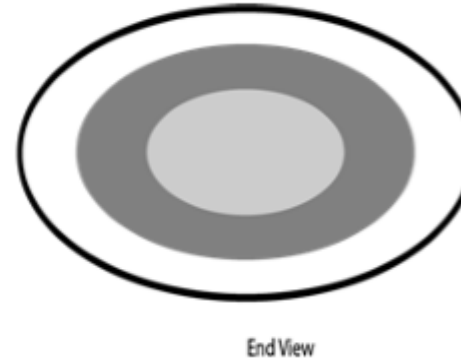
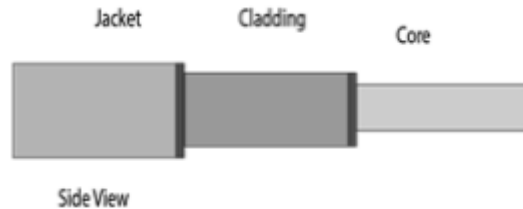
- It is more expensive as compared to twisted pair cable.
- If any fault occurs in the cable causes the failure in the entire network.

Fiber Optic Cable

- Fiber optic cable is a cable that uses light signals for communication.
- Fiber optic is a cable that holds the optical fibers coated in plastic that are used to send the data by pulses of light.
- The plastic coating protects the optical fibers from heat, cold, electromagnetic interference from other types of wiring.
- Fiber optics provide faster data transmission than copper wires.



Fiber Optic – Views & Modes

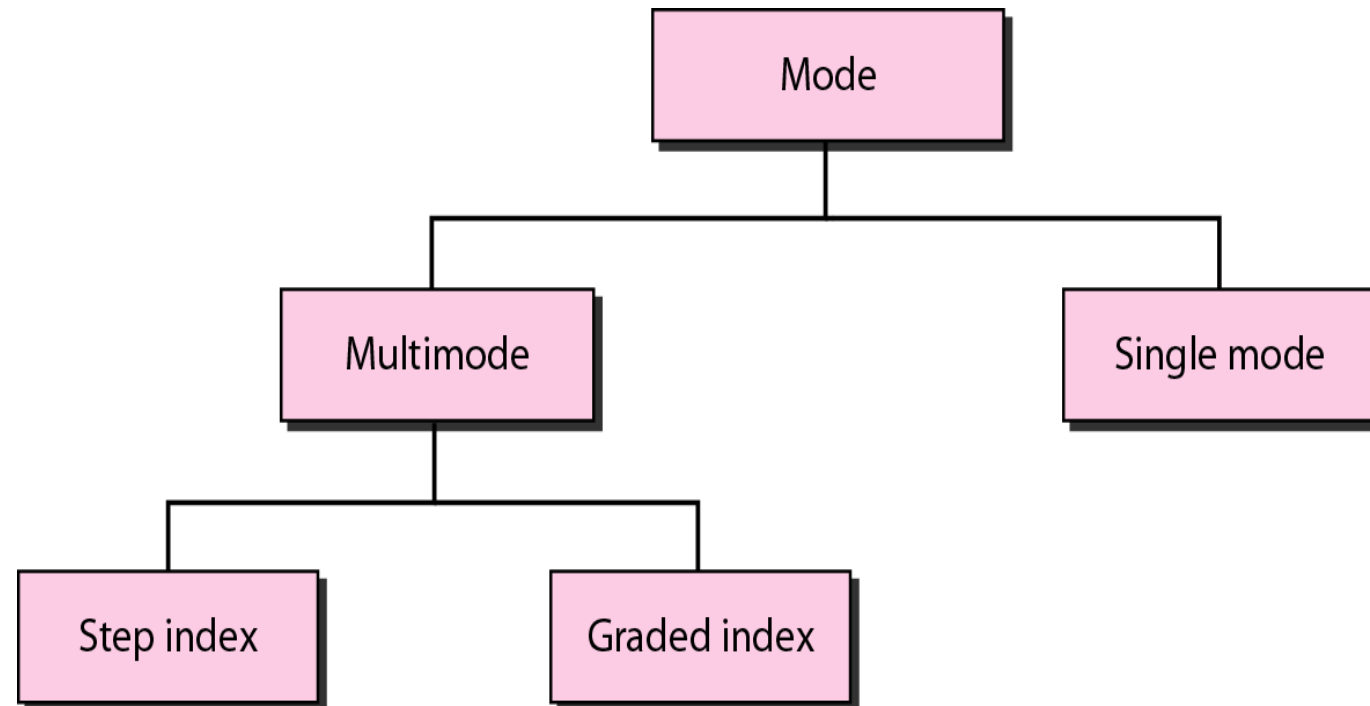


Basic elements of Fiber optic cable:

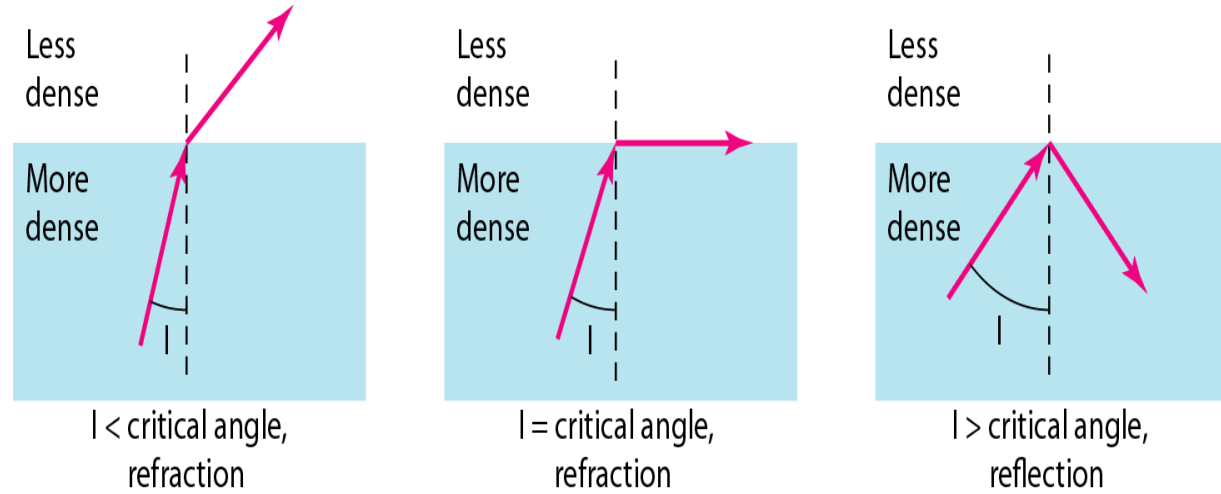
Core: The optical fiber consists of a narrow strand of glass or plastic known as a core. A core is a light transmission area of the fiber. The more the area of the core, the more light will be transmitted into the fiber.

Cladding: The concentric layer of glass is known as cladding. The main functionality of the cladding is to provide the lower refractive index at the core interface as to cause the reflection within the core so that the light waves are transmitted through the fiber.

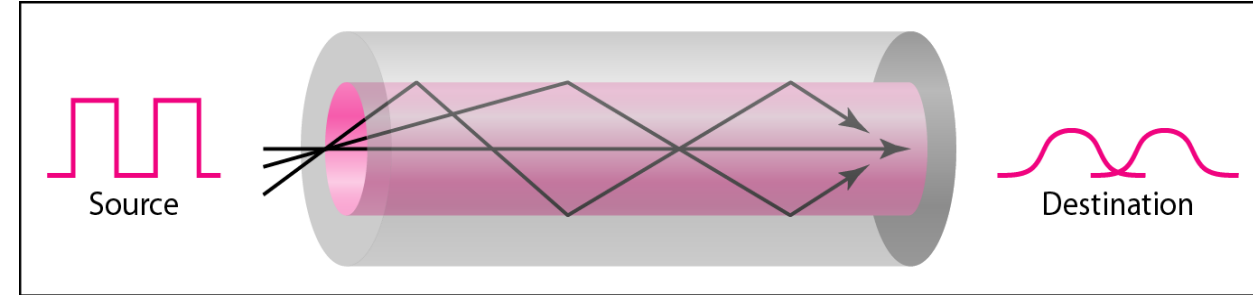
Jacket: The protective coating consisting of plastic is known as a jacket. The main purpose of a jacket is to preserve the fibre strength, absorb shock and extra fiber protection.



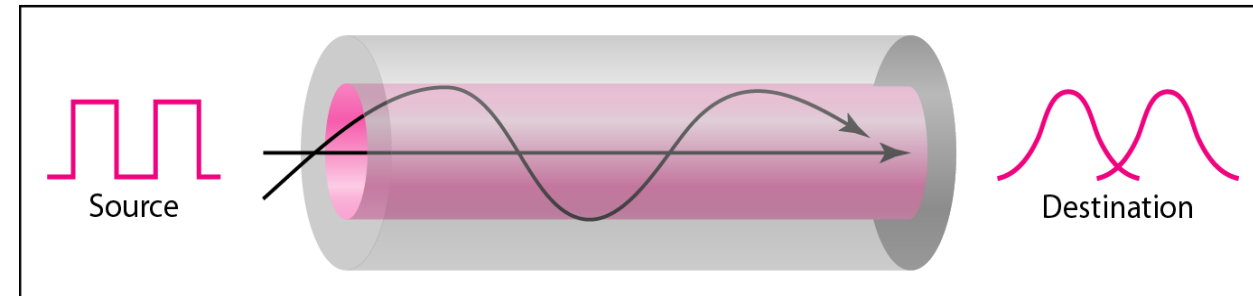
Fiber Optic – Modes



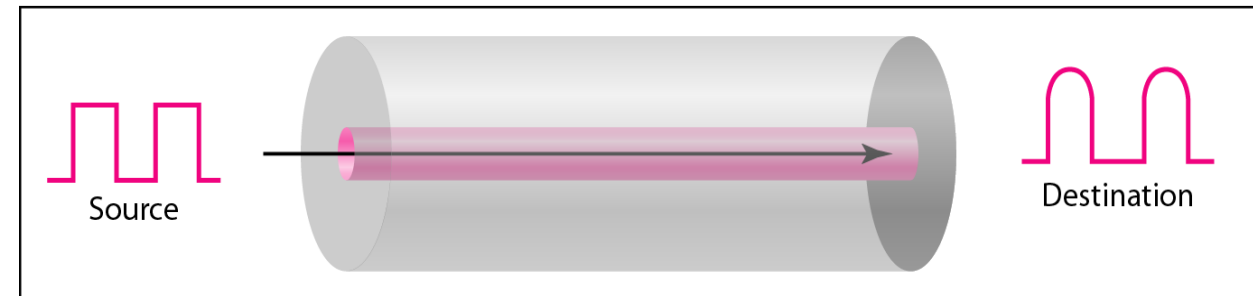
Type	Core (μm)	Cladding (μm)	Mode
50/125	50.0	125	Multimode, graded index
62.5/125	62.5	125	Multimode, graded index
100/125	100.0	125	Multimode, graded index
7/125	7.0	125	Single mode



a. Multimode, step index



b. Multimode, graded index



c. Single mode

Advantages & Disadvantages of OFC

Advantages of Optical Fiber Cable

- Increased capacity and bandwidth
- Lightweight
- Less signal attenuation
- Immunity to electromagnetic interference
- Resistance to corrosive materials

Disadvantages of Optical Fiber Cable

- Difficult to install and maintain
- High cost
- Fragile

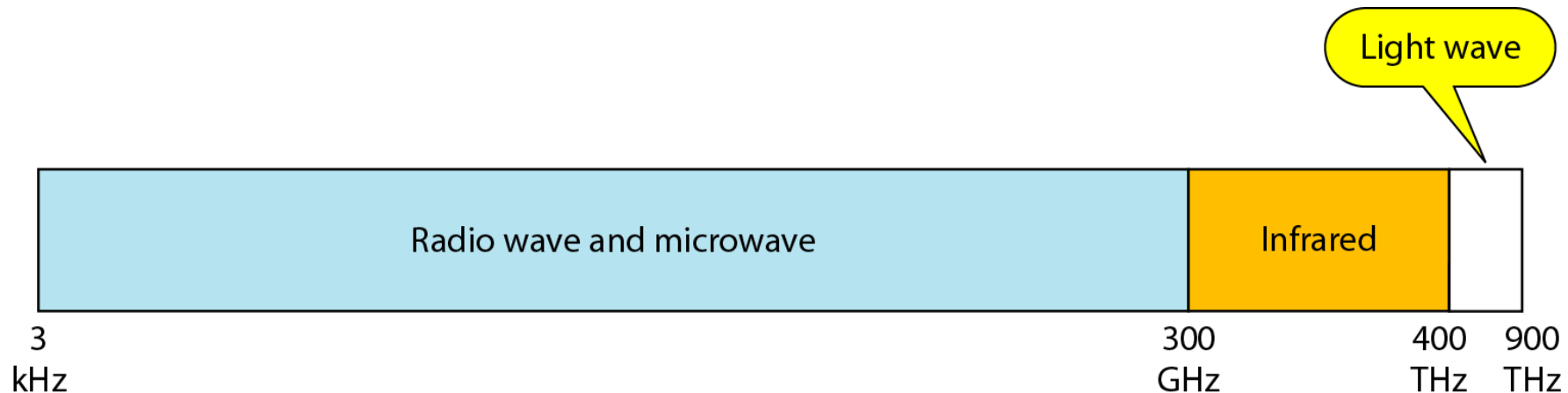
UNGUIDED MEDIA

- *Unguided media transport electromagnetic waves without using a physical conductor. This type of communication is often referred to as wireless communication.*

Radio Waves

Microwaves

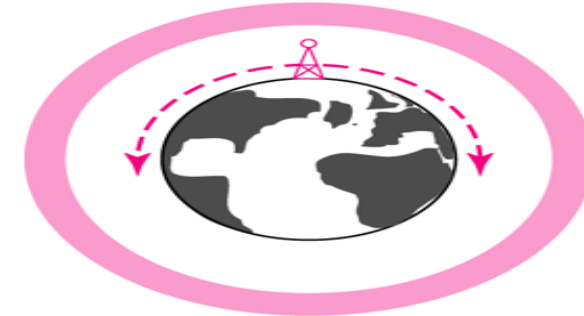
Infrared



Frequency Bands

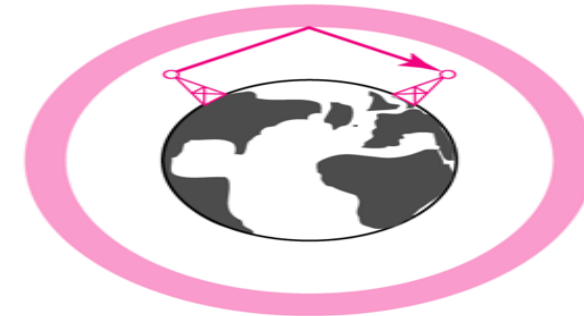
<i>Band</i>	<i>Range</i>	<i>Propagation</i>	<i>Application</i>
VLF (very low frequency)	3–30 kHz	Ground	Long-range radio navigation
LF (low frequency)	30–300 kHz	Ground	Radio beacons and navigational locators
MF (middle frequency)	300 kHz–3 MHz	Sky	AM radio
HF (high frequency)	3–30 MHz	Sky	Citizens band (CB), ship/aircraft communication
VHF (very high frequency)	30–300 MHz	Sky and line-of-sight	VHF TV, FM radio
UHF (ultrahigh frequency)	300 MHz–3 GHz	Line-of-sight	UHF TV, cellular phones, paging, satellite
SHF (superhigh frequency)	3–30 GHz	Line-of-sight	Satellite communication
EHF (extremely high frequency)	30–300 GHz	Line-of-sight	Radar, satellite

Ionosphere



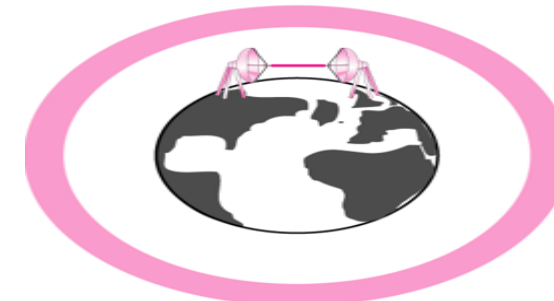
Ground propagation
(below 2 MHz)

Ionosphere



Sky propagation
(2–30 MHz)

Ionosphere



Line-of-sight propagation
(above 30 MHz)

Comparison

Basis	Radiowave	Microwave	Infrared wave
Direction	These are omni-directional in nature.	These are unidirectional in nature.	These are unidirectional in nature.
Penetration	At low frequency, they can penetrate through solid objects and walls but high frequency they bounce off the obstacle.	At low frequency, they can penetrate through solid objects and walls. at high frequency, they cannot penetrate.	They cannot penetrate through any solid object and walls.
Frequency range	Frequency range: 3 KHz to 1GHz.	Frequency range: 1 GHz to 300 GHz.	Frequency range: 300 GHz to 400 GHz.
Security	These offers poor security.	These offers medium security.	These offers high security.
Attenuation	Attenuation is high.	Attenuation is variable.	Attenuation is low.
Government License	Some frequencies in the radio-waves require government license to use these.	Some frequencies in the microwaves require government license to use these.	There is no need of government license to use these waves.
Usage Cost	Setup and usage Cost is moderate.	Setup and usage Cost is high.	Usage Cost is very less.
Communication	These are used in long distance communication.	These are used in long distance communication.	These are not used in long distance communication.

Summary

Topics discussed in Unit-I

- Network Types – LAN, MAN, WAN, PAN
- Network Topology – BUS, STAR, RING, MESH & HYBRID
- Switching – Circuit & Packet switching
- OSI Layered Architecture
- TCP/IP Model
- Physical Layer Overview – Latency, B/W etc.
- Guided Media – UTP, STP, Coaxial & Fiber Optic
- Unguided Media – Radio waves, Microwave & Infrared